MRI in Neuroscience and Neurology · WiSe 2026/27 · session 6

Brain region segmentations using FSL, FreeSurfer and Python

Lecture of 18 November 2026 · 50 slides

Grey matter (FreeSurfer, SynthSeg) · white matter lesions (BIANCA, U-Net, Dice) · microbleeds (U-Net, F1)

Online deck with voice, clips and the interactive pages:
mendeltem.github.io/mri-segmentation-lectures/lecture_1118/deck/

All example images come from public data sets; see the sources of each slide.

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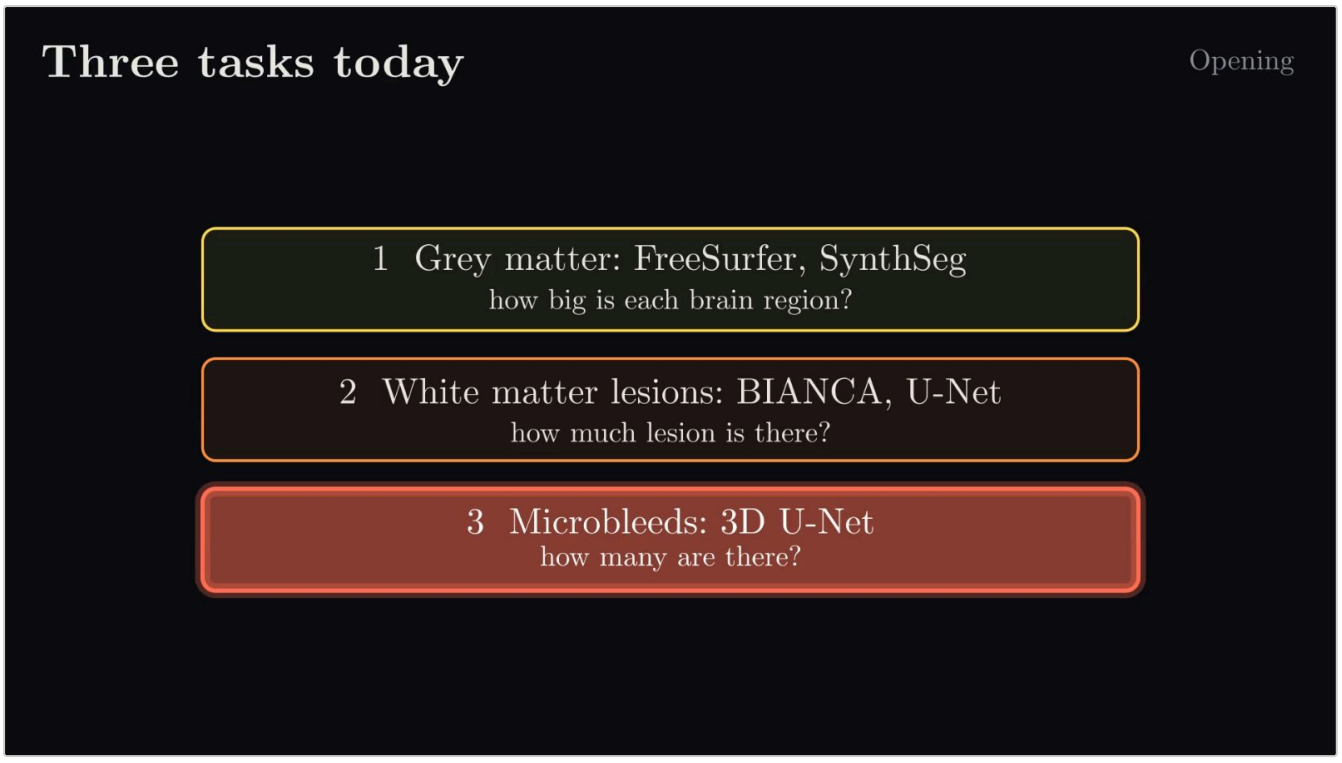
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All sources (37)

1. Three tasks today



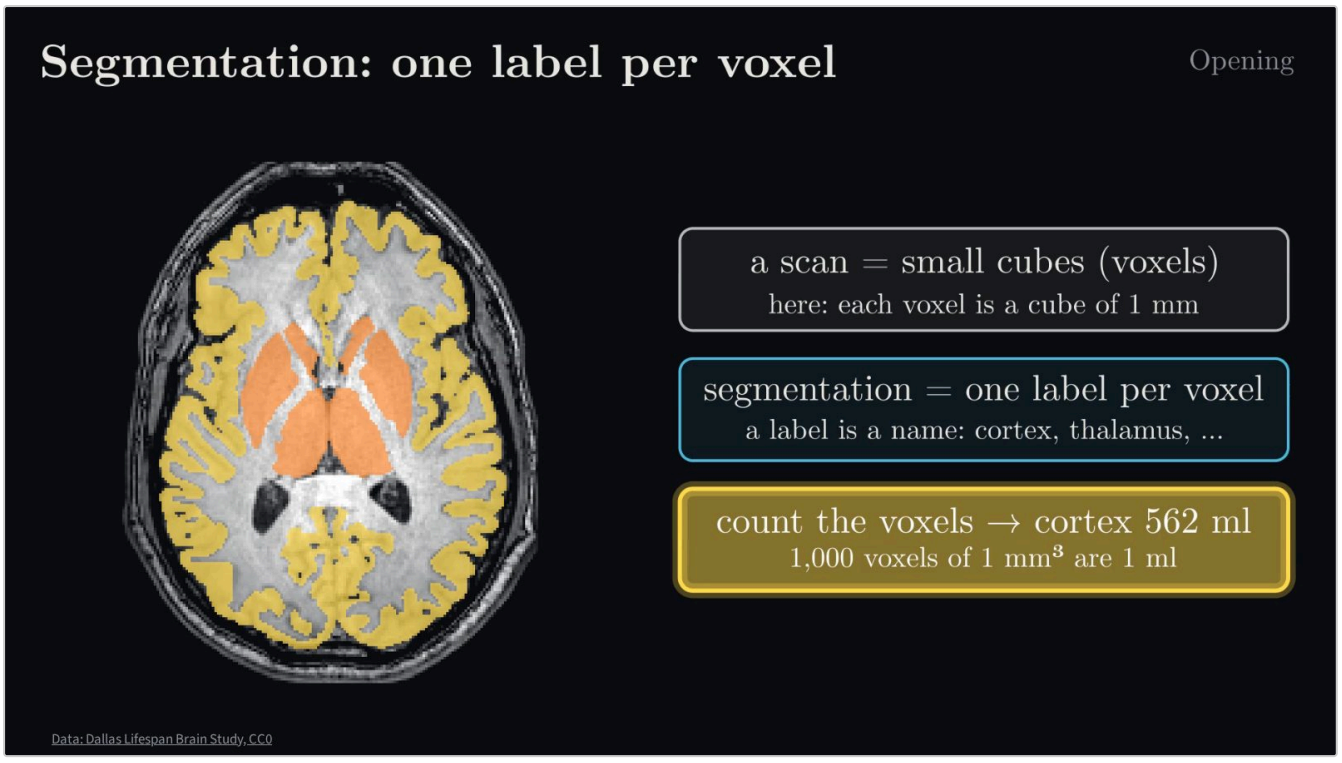
KEY POINTS

- **1 Grey matter: FreeSurfer, SynthSeg** – how big is each brain region?
- **2 White matter lesions: BIANCA, U-Net** – how much lesion is there?
- **3 Microbleeds: 3D U-Net** – how many are there?

WHAT IS SAID

Three tasks today. Grey matter: how big is each brain region? White matter lesions: how much is there? And microbleeds: how many are there?

2. Segmentation: one label per voxel



KEY POINTS

- **a scan = small cubes (voxels)** – here: each voxel is a cube of 1 mm
- **segmentation = one label per voxel** – a label is a name: cortex, thalamus, …
- **count the voxels → cortex 562 ml** – 1,000 voxels of 1 mm³ are 1 ml

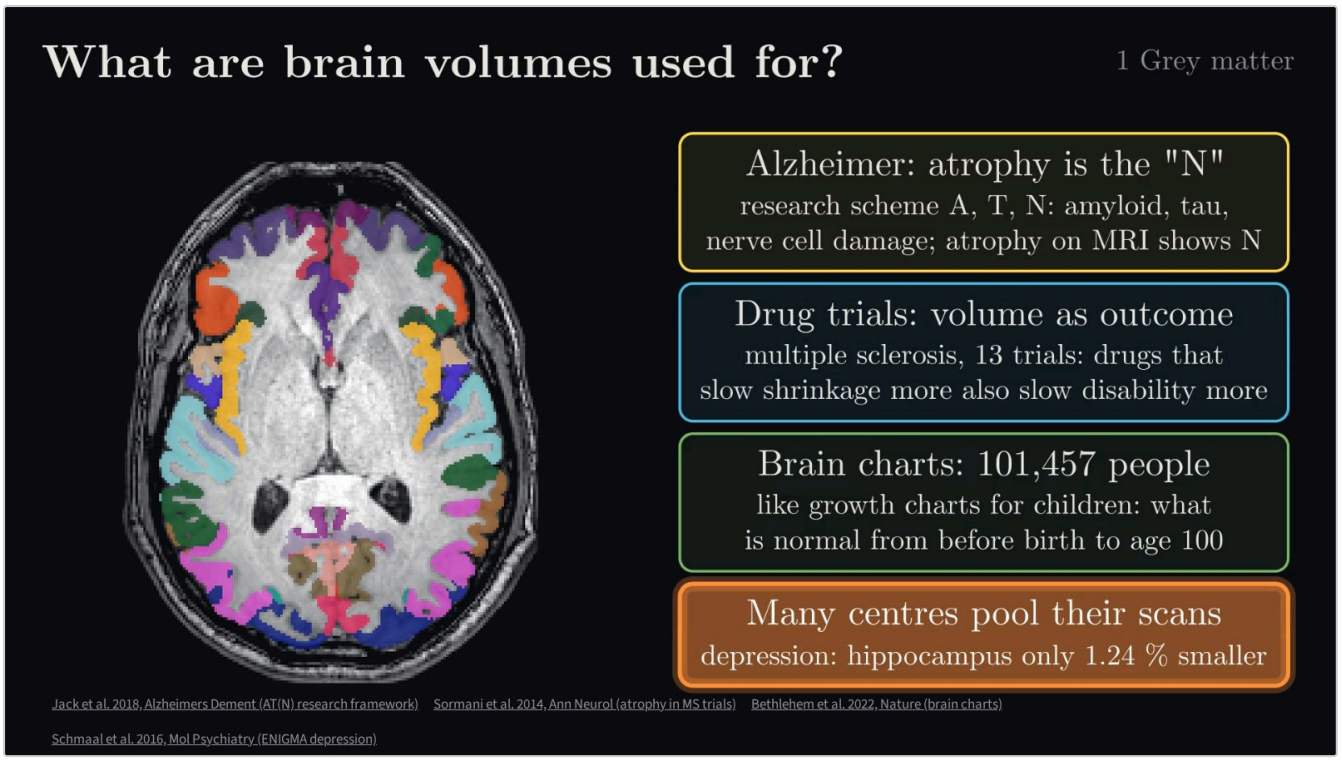
WHAT IS SAID

An MRI scan is made of small cubes, the voxels. Here, each one is a cube of one millimetre. Segmentation gives every voxel a label: a name, like cortex or thalamus. Count the voxels of one label, and you have its volume. A thousand voxels are one millilitre. This cortex: five hundred sixty two millilitres.

SOURCES

- Data: Dallas Lifespan Brain Study, CC0
  Dallas Lifespan Brain Study: public MRI data (CC0), used for our grey-matter pictures.

3. What are brain volumes used for?



KEY POINTS

- **Alzheimer: atrophy is the "N"** – research scheme A, T, N: amyloid, tau, nerve cell damage; atrophy on MRI shows N
- **Drug trials: volume as outcome** – multiple sclerosis, 13 trials: drugs that slow shrinkage more also slow disability more
- **Brain charts: 101,457 people** – like growth charts for children: what is normal from before birth to age 100
- **Many centres pool their scans** – depression: hippocampus only 1.24 % smaller

WHAT IS SAID

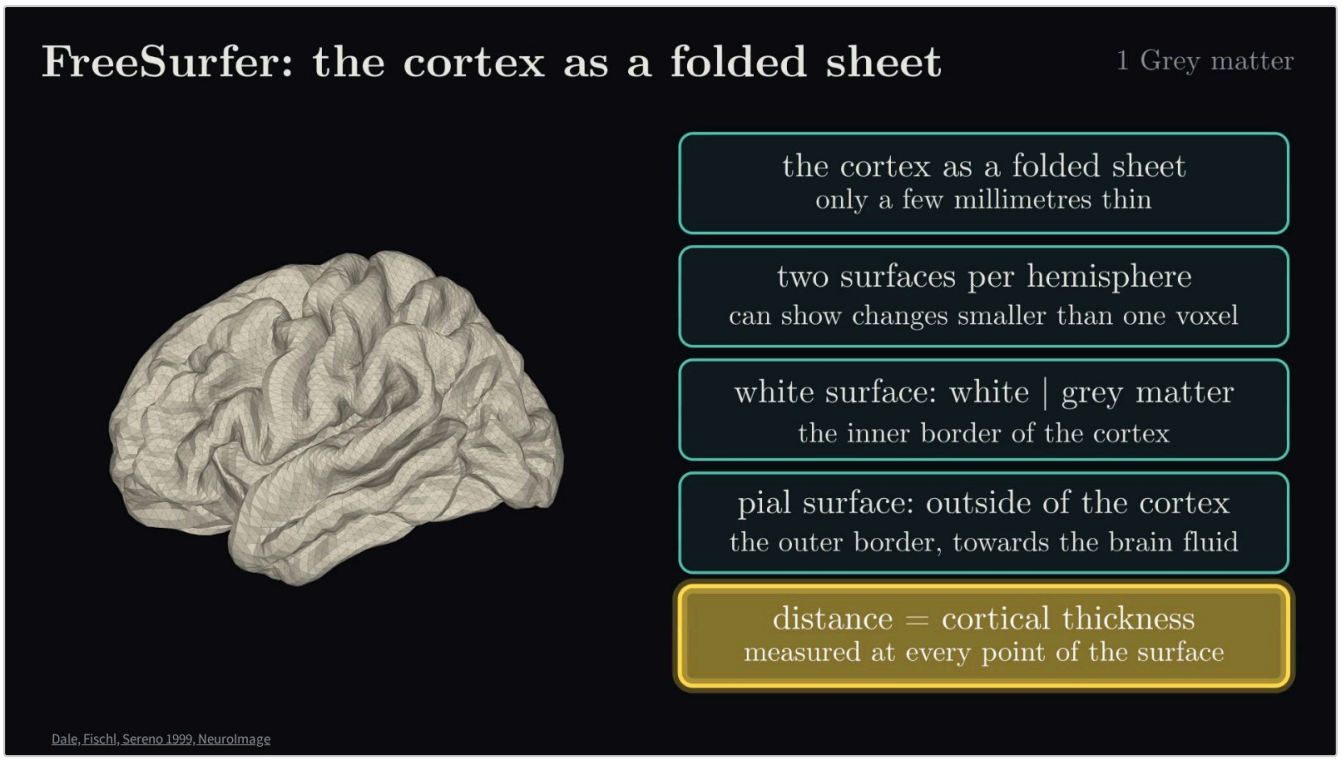
Why measure volumes at all? Three real examples. In Alzheimer research, atrophy, the shrinking of the brain, is one of three markers of the disease. Drug trials use it as an outcome. In multiple sclerosis, drugs that slowed the shrinking more also slowed disability more. Brain charts pool more than a hundred thousand people. Like growth charts for children, they show what is normal at each age. And many centres pool their scans. In depression, the hippocampus is about one percent smaller: too little to see in one patient, but clear in thousands.

SOURCES

- Jack et al. 2018, Alzheimers Dement (AT(N) research framework)
  Research definition of Alzheimer's disease by biomarkers: amyloid (A), tau (T), neurodegeneration (N). Atrophy on MRI is one (N) marker; not specific, and not meant for routine practice.
- Sormani et al. 2014, Ann Neurol (atrophy in MS trials)
  Meta-analysis of 13 multiple sclerosis trials (more than 13,500 patients): the more a drug slowed brain atrophy, the more it slowed disability (R2 = 0.48).

- Bethlehem et al. 2022, Nature (brain charts)
  Brain charts for the human lifespan: 123,984 scans of 101,457 people, from 115 days after conception to 100 years; gives centiles like a growth chart.
- Schmaal et al. 2016, Mol Psychiatry (ENIGMA depression)
  ENIGMA consortium: 1,728 patients with depression and 7,199 controls from 15 samples; hippocampus 1.24 % smaller (Cohen's d = -0.14).

4. FreeSurfer: the cortex as a folded sheet



KEY POINTS

- **the cortex as a folded sheet** – only a few millimetres thin
- **two surfaces per hemisphere** – can show changes smaller than one voxel
- **white surface: white | grey matter** – the inner border of the cortex
- **pial surface: outside of the cortex** – the outer border, towards the brain fluid
- **distance = cortical thickness** – measured at every point of the surface

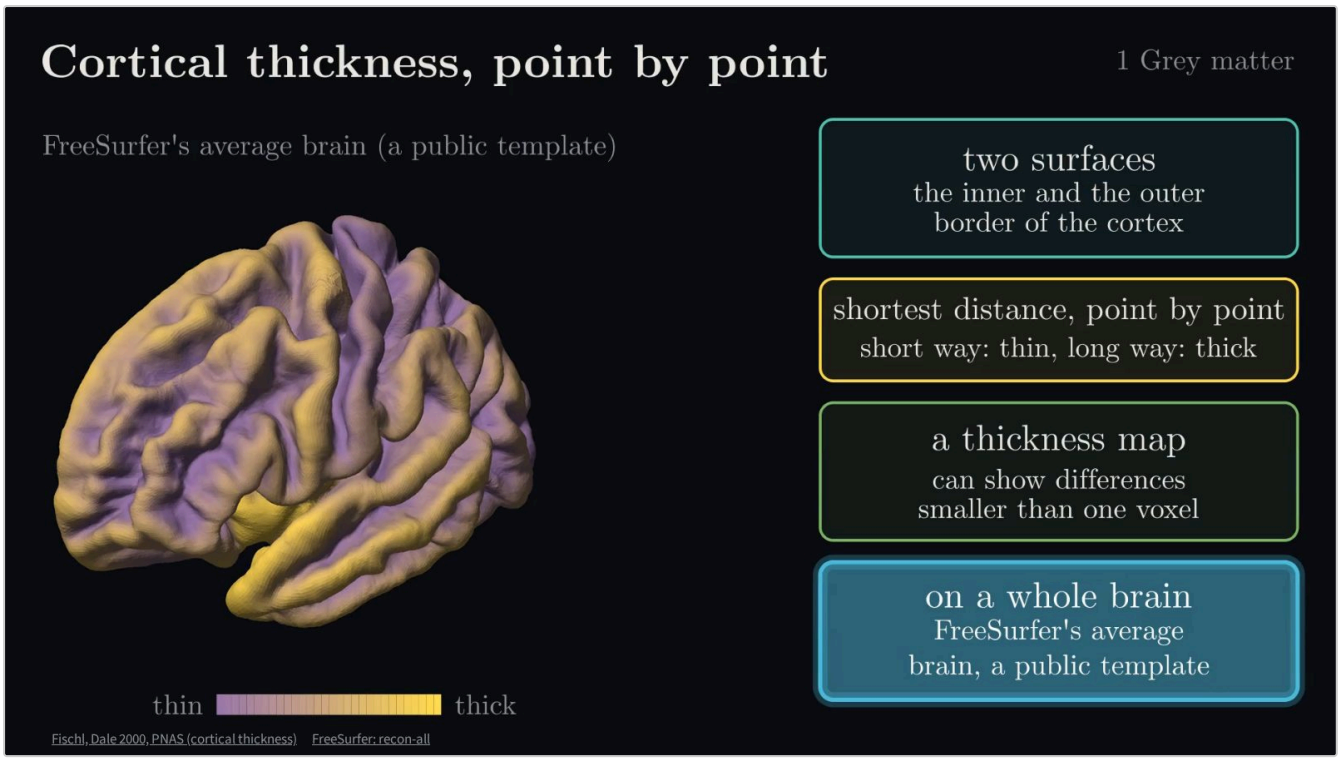
WHAT IS SAID

FreeSurfer is the classic tool. It sees the cortex as a folded sheet, only a few millimetres thin. So it does not count voxels: it draws two surfaces. The white surface is the inner border, between white and grey matter. The pial surface is the outer border. The distance between them is the cortical thickness.

SOURCES

- Dale, Fischl, Sereno 1999, NeuroImage
  FreeSurfer part I: segmentation and the white and pial surface of the cortex.

5. Cortical thickness, point by point



KEY POINTS

- **two surfaces** – the inner and the outer border of the cortex
- **pial surface** – the outer border
- **shortest distance, point by point**
- **a thickness map** – can show differences smaller than one voxel
- **on a whole brain** – FreeSurfer's average brain, a public template

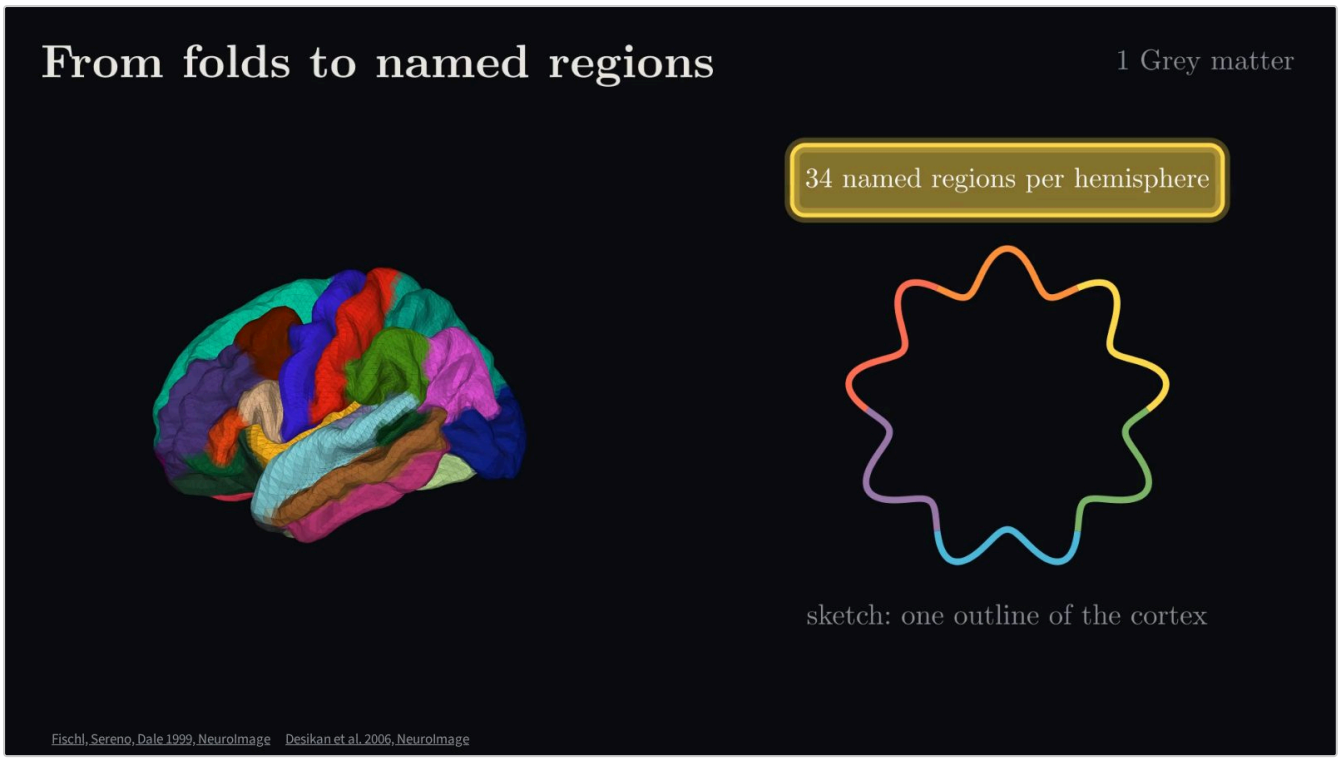
WHAT IS SAID

How thick is the cortex at one point? FreeSurfer has two surfaces. The white surface is the inner border. The pial surface is the outer border. From every point of one surface, it measures the shortest way to the other surface. A short way means thin cortex. A long way means thick cortex. Colour every point by its distance, and you get a thickness map. The surfaces run between the voxels. So the map can show differences smaller than one voxel. You see where the cortex gets thinner, not only that it shrinks. Now the same on a whole brain: FreeSurfer's average brain. The white surface grows outwards, to the pial surface. Then every point gets its colour. Purple is thin cortex. Yellow is thick cortex.

SOURCES

- Fischl, Dale 2000, PNAS (cortical thickness)
  How FreeSurfer measures cortical thickness from the two surfaces; differences between people below 0.5 mm.
- FreeSurfer: recon-all
  FreeSurfer documentation of recon-all, the command that runs the whole pipeline.

6. From folds to named regions



KEY POINTS

- **inflate the surface**
- **… then a sphere**
- **match the folds to an average brain**
- **every fold gets a name: 34 regions per hemisphere**

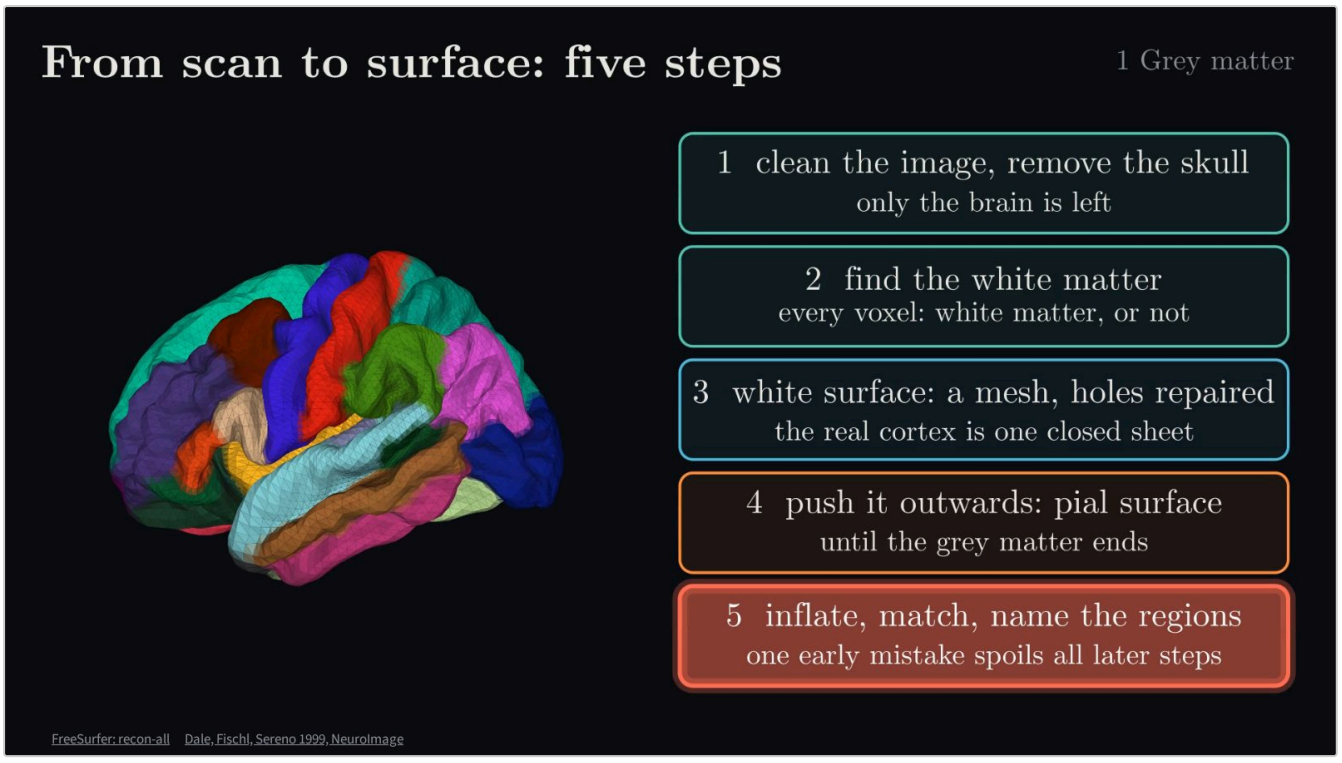
WHAT IS SAID

To name the folds, FreeSurfer blows the surface up into a sphere. There it matches the folds to an average brain. So every fold gets a name: thirty four regions per hemisphere.

SOURCES

- Fischl, Sereno, Dale 1999, NeuroImage
  FreeSurfer part II: the cortex inflated and mapped to a sphere; folding patterns aligned between brains.
- Desikan et al. 2006, NeuroImage
  Automatic labelling of 34 cortical regions per hemisphere, learned from 40 hand-labelled brains.

7. From scan to surface: five steps



KEY POINTS

- **1 clean the image, remove the skull** – only the brain is left
- **2 find the white matter** – every voxel: white matter, or not
- **3 white surface: a mesh, holes repaired** – the real cortex is one closed sheet
- **4 push it outwards: pial surface** – until the grey matter ends
- **5 inflate, match, name the regions** – one early mistake spoils all later steps

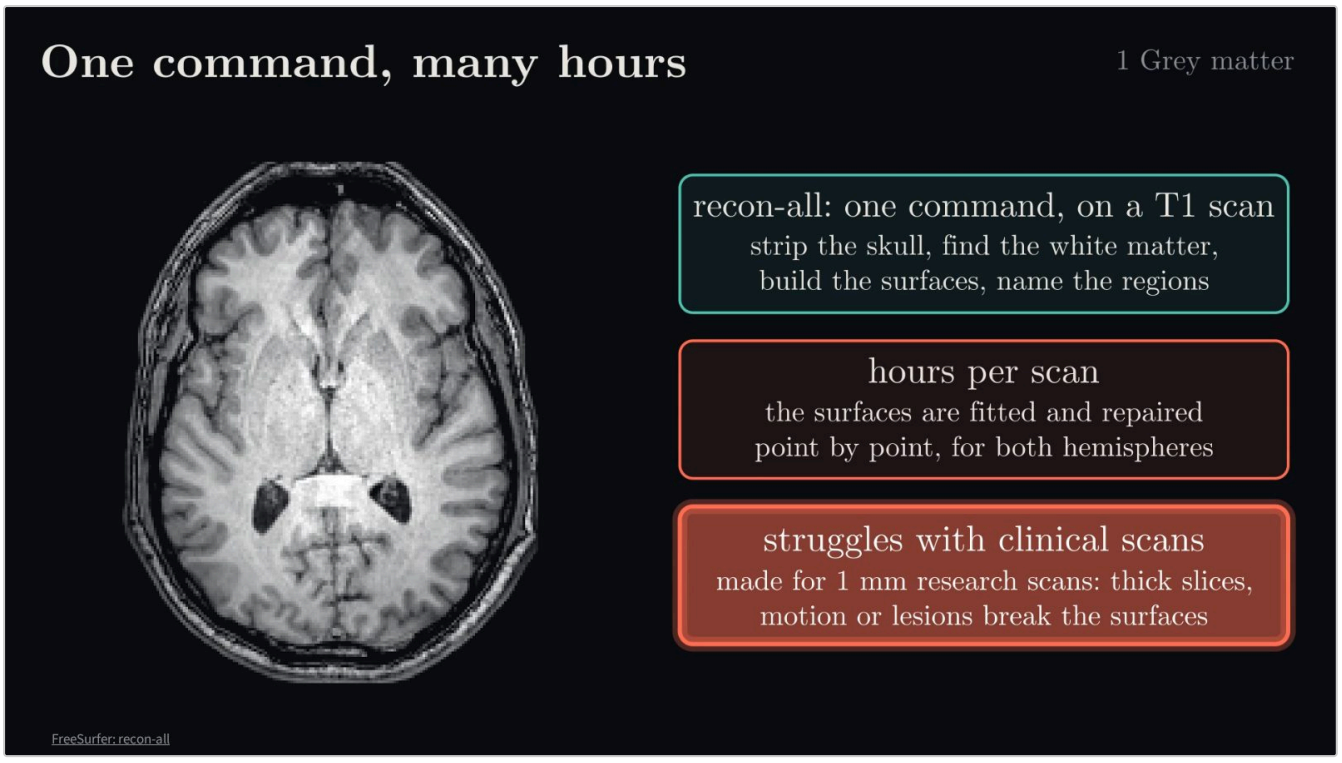
WHAT IS SAID

How does FreeSurfer get from a scan to these surfaces? In a fixed chain of steps. First it cleans the image and removes the skull. Only the brain is left. Then every voxel is sorted: white matter, or not. The white matter becomes one solid body in each hemisphere. Around this body it lays a mesh of small triangles: the white surface. Holes are repaired, because the real cortex is one closed sheet. This surface is pushed outwards, until the grey matter ends. That is the pial surface. Last, the surface is inflated, matched to the average brain, and the regions get their names. Every step needs the one before. So one early mistake, such as a piece of skull left behind, spoils everything after it.

SOURCES

- FreeSurfer: recon-all
  FreeSurfer documentation of recon-all, the command that runs the whole pipeline.
- Dale, Fischl, Sereno 1999, NeuroImage
  FreeSurfer part I: segmentation and the white and pial surface of the cortex.

8. One command, many hours



KEY POINTS

- **recon-all: one command, on a T1 scan** – strip the skull, find the white matter, build the surfaces, name the regions
- **hours per scan** – the surfaces are fitted and repaired point by point, for both hemispheres
- **struggles with clinical scans** – made for 1 mm research scans: thick slices, motion or lesions break the surfaces

WHAT IS SAID

One command does it all: recon all, on a T one scan. Behind it are many steps, one after the other. It needs time: hours per scan, because the surfaces are fitted and repaired point by point. And it struggles with clinical scans. It was made for research scans, with thin slices and good contrast. Thick slices, motion or lesions break the surfaces.

SOURCES

- FreeSurfer: recon-all
  FreeSurfer documentation of recon-all, the command that runs the whole pipeline.

9. SynthSeg: trained on images that never existed



KEY POINTS

- **a neural network that never saw a real scan**
- **trained on fake images: random contrast**
- **a real scan = one more variation**

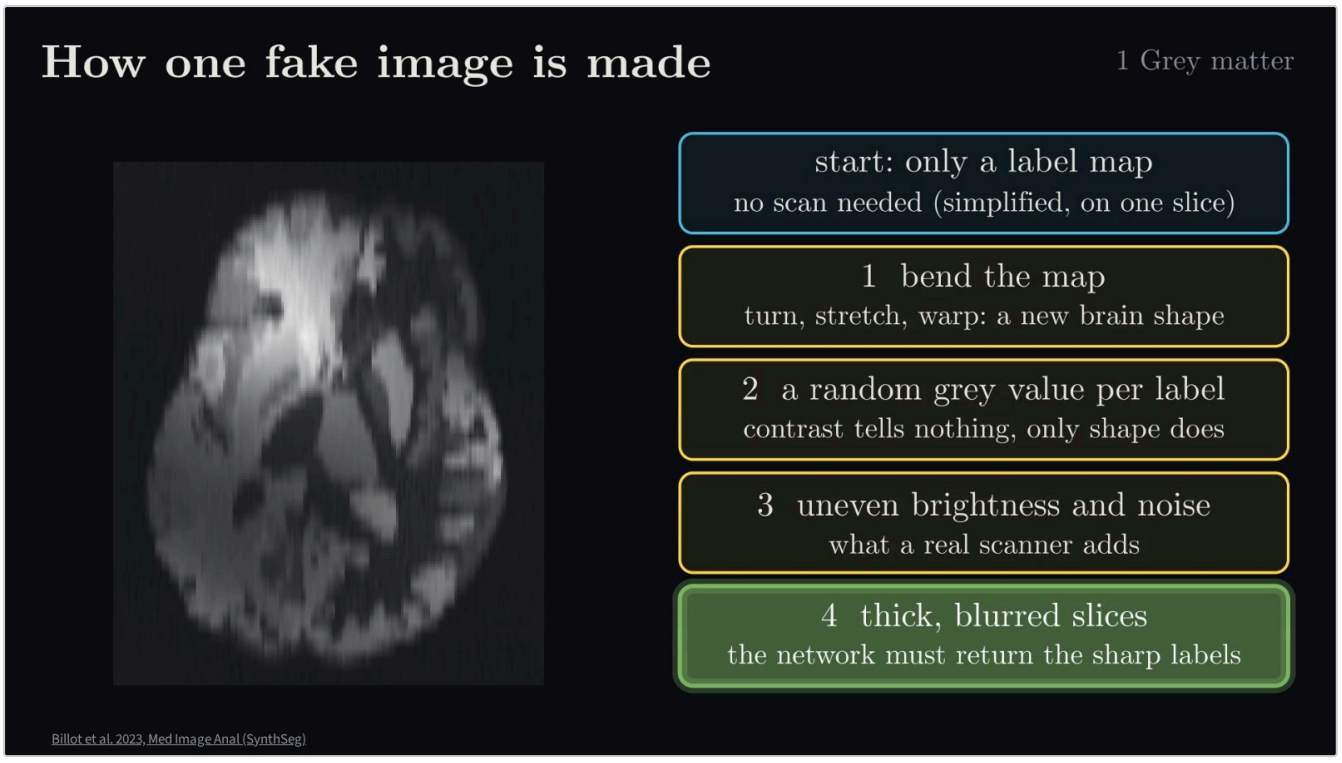
WHAT IS SAID

SynthSeg is a neural network that never saw a real scan. It trained on fake images with random contrast and resolution. So a real scan is just one more variation.

SOURCES

- Billot et al. 2023, Med Image Anal (SynthSeg)
  SynthSeg: a network trained only on synthetic images; segments scans of any contrast and resolution.

10. How one fake image is made



KEY POINTS

- **start: only a label map** – no scan needed (simplified, on one slice)
- **1 bend the map** – turn, stretch, warp: a new brain shape
- **2 a random grey value per label** – contrast tells nothing, only shape does
- **3 uneven brightness and noise** – what a real scanner adds
- **4 thick, blurred slices** – the network must return the sharp labels

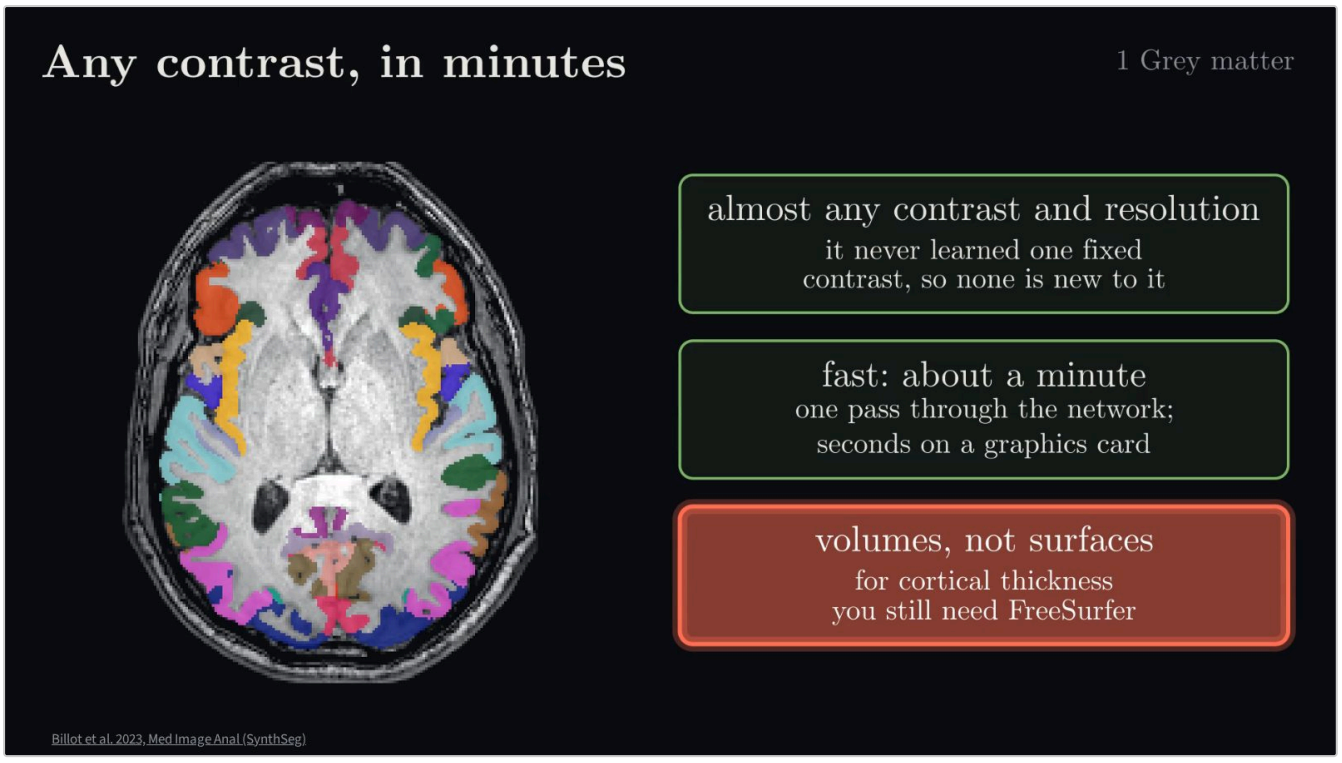
WHAT IS SAID

How is one fake image made? It starts with a label map of a brain. No scan, only the labels. First, bend it. The map is turned, stretched and warped at random. So every brain shape is new. Second, paint it. Every label gets a random grey value. White matter may be bright, dark, or in between. That is the trick: the contrast tells the network nothing. Only the shape does. Third, add what a scanner adds: uneven brightness across the image, and noise. Fourth, make it coarse: thick, blurred slices, like a clinical scan. The network gets this image, and must give back the sharp label map. And at every training step, it gets a new one.

SOURCES

- Billot et al. 2023, Med Image Anal (SynthSeg)
  SynthSeg: a network trained only on synthetic images; segments scans of any contrast and resolution.

11. Any contrast, in minutes



KEY POINTS

- **almost any contrast and resolution** – it never learned one fixed contrast, so none is new to it
- **fast: about a minute** – one pass through the network; seconds on a graphics card
- **volumes, not surfaces** – for cortical thickness you still need FreeSurfer

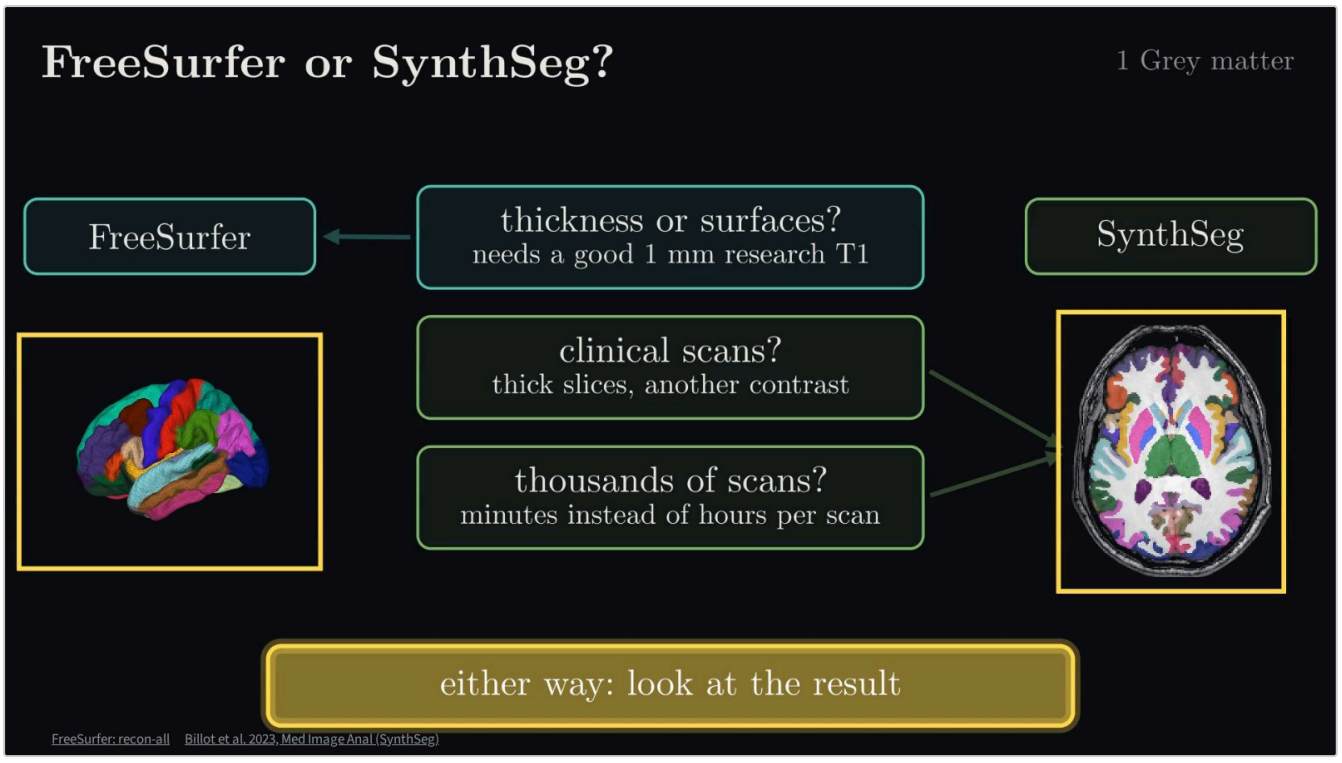
WHAT IS SAID

Because of this training, it works on almost any contrast and resolution, without retraining. And it is fast: the scan goes through the network once. About a minute on a normal computer. But it gives volumes, not surfaces. For cortical thickness, you still need FreeSurfer.

SOURCES

- Billot et al. 2023, Med Image Anal (SynthSeg)
  SynthSeg: a network trained only on synthetic images; segments scans of any contrast and resolution.

12. FreeSurfer or SynthSeg?



KEY POINTS

- **thickness or surfaces? FreeSurfer** – needs a good 1 mm research T1
- **clinical scans? SynthSeg** – thick slices, another contrast
- **thousands of scans? SynthSeg** – minutes instead of hours per scan
- **either way: look at the result**

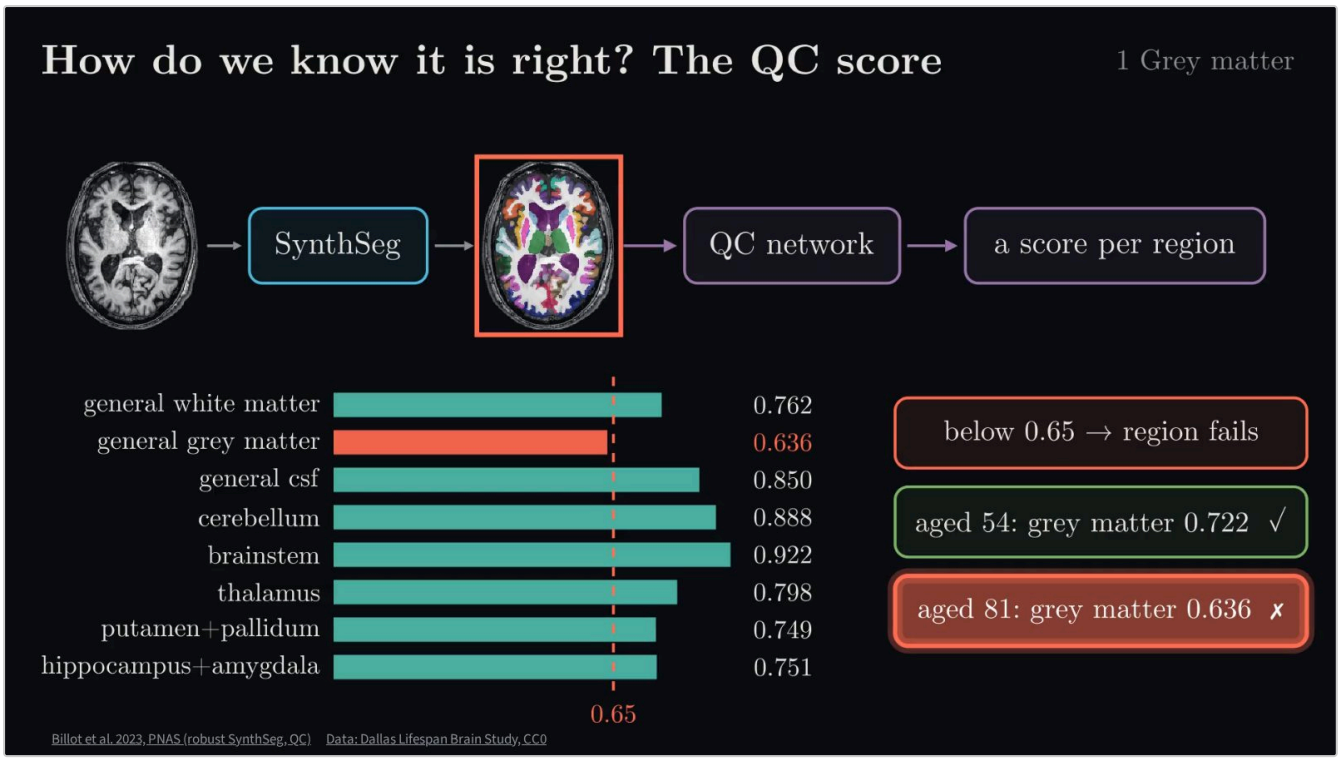
WHAT IS SAID

FreeSurfer or SynthSeg? Ask three questions. One: do you need cortical thickness, or surfaces? Then FreeSurfer. And then you need a good research scan. Two: are your scans clinical, with thick slices or another contrast? Then SynthSeg. Three: do you have thousands of scans, and little time? Also SynthSeg. And whichever you choose: look at the result. Both tools always give an answer. Not always a right one.

SOURCES

- FreeSurfer: recon-all
  FreeSurfer documentation of recon-all, the command that runs the whole pipeline.
- Billot et al. 2023, Med Image Anal (SynthSeg)
  SynthSeg: a network trained only on synthetic images; segments scans of any contrast and resolution.

13. How do we know it is right? The QC score



KEY POINTS

- **a second small network gives the score**
- **it sees only the label map, not the scan**
- **it has learned: good and broken label maps**
- **it guesses the quality: a score from 0 to 1**
- **below 0.65 → region fails**
- **aged 54: grey matter 0.722 ✓**
- **aged 81: grey matter 0.636 ✗**

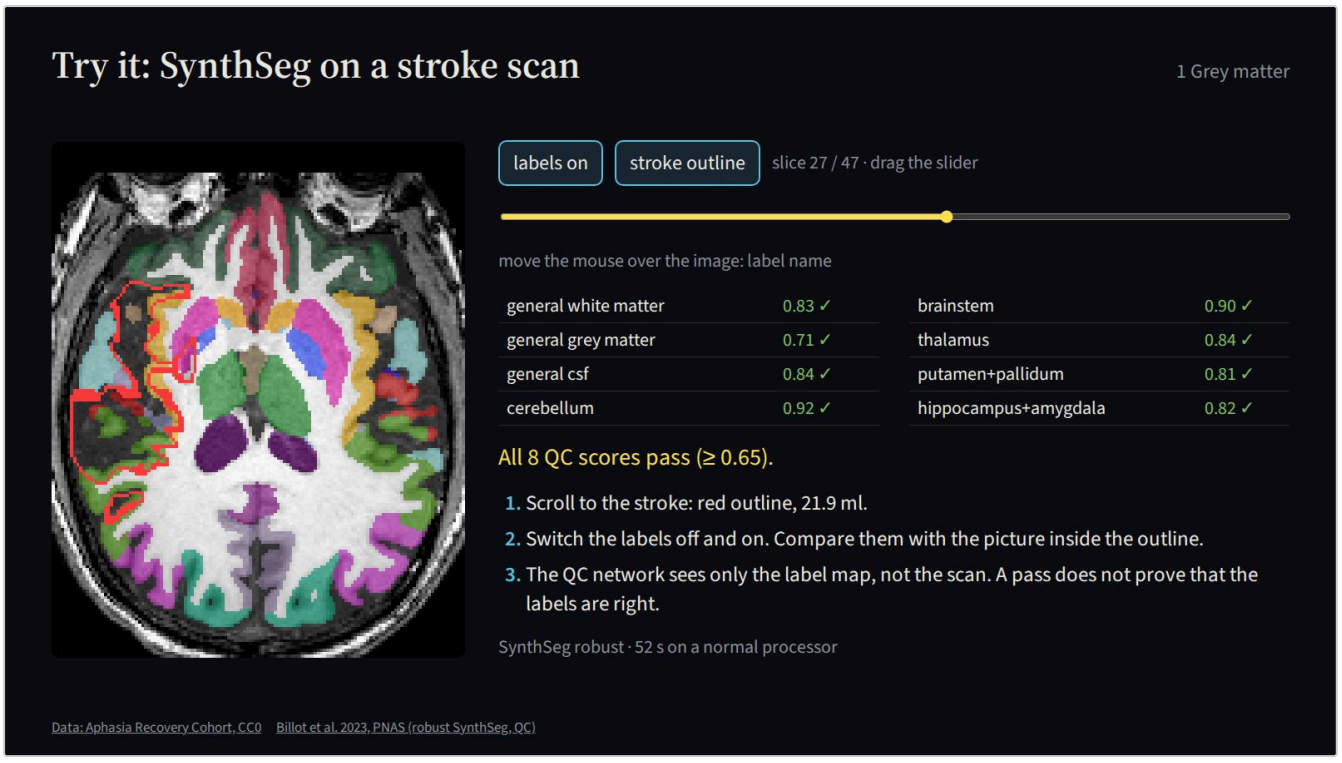
WHAT IS SAID

How do we know a segmentation is right? Nobody can look at ten thousand scans. So SynthSeg scores its own result. A second small network looks only at the label map, not at the scan. It has learned how good label maps look, and how broken ones look. So for each region, it guesses the quality: a score between zero and one. Below zero point six five, a region fails. This scan passes. This one fails. Look at it.

SOURCES

- Billot et al. 2023, PNAS (robust SynthSeg, QC)
  SynthSeg+ (robust version) with automatic QC scores per region; below 0.65 a region counts as failed.
- Data: Dallas Lifespan Brain Study, CC0
  Dallas Lifespan Brain Study: public MRI data (CC0), used for our grey-matter pictures.

14. Try it: SynthSeg on a stroke scan



> Interactive page: try it in the online deck – mendeltem.github.io/mri-segmentation-lectures/lecture_1118/deck/#try_qc

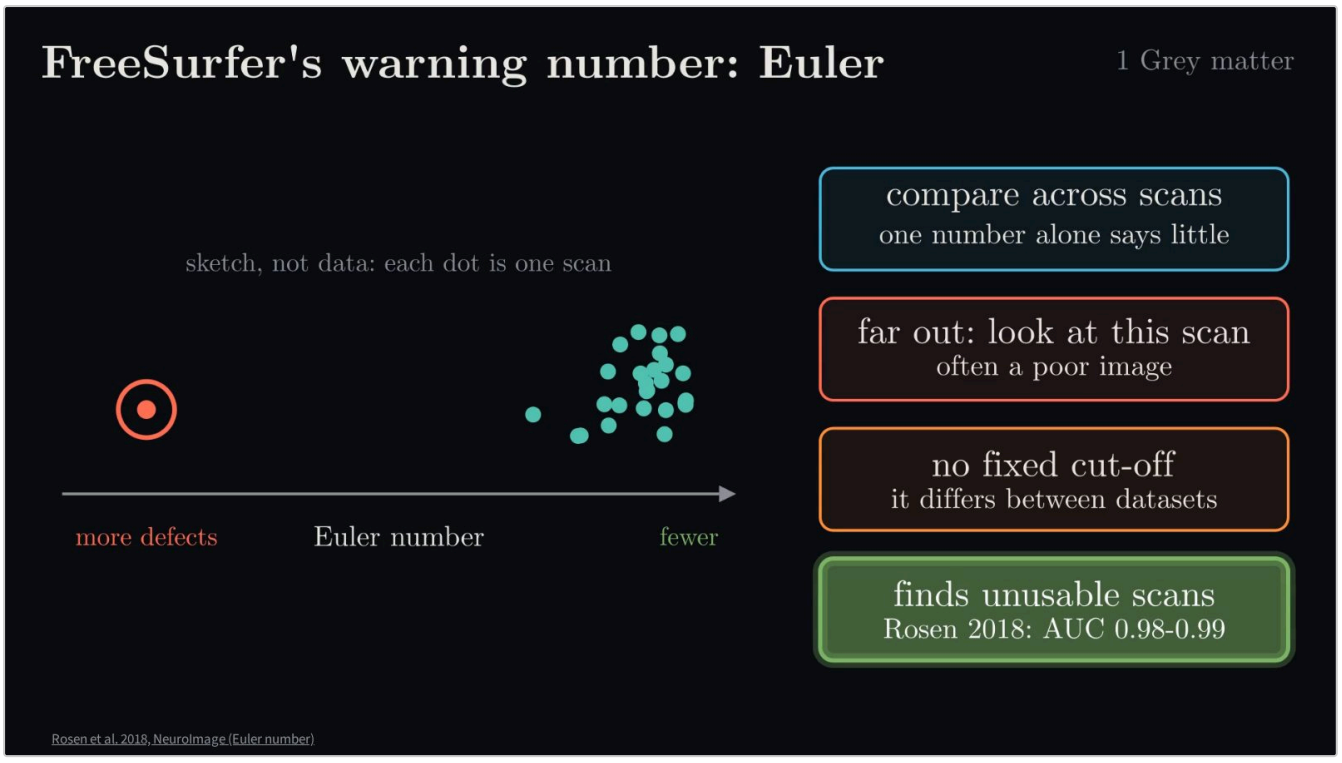
WHAT IS SAID

Scroll through a stroke scan with SynthSeg's labels. All QC values pass, yet the labels around the lesion are not right. A passing QC score is necessary, not sufficient. SynthSeg took 52 seconds on a normal processor.

SOURCES

- Data: Aphasia Recovery Cohort, CC0
  Aphasia Recovery Cohort: public stroke MRI with expert lesion masks (CC0).
- Billot et al. 2023, PNAS (robust SynthSeg, QC)
  SynthSeg+ (robust version) with automatic QC scores per region; below 0.65 a region counts as failed.

15. FreeSurfer's warning number: Euler



KEY POINTS

- **corners − edges + faces** – count them on any surface made of flat pieces
- **closed, no holes: always 2** – a cube, a ball, a perfect cortex surface
- **each handle: minus 2** – a handle is like the handle of a cup
- **first surface: 26 defects** – public stroke scan, left side
- **Euler number −38** – after the repair: 2
- **compare across scans** – one number alone says little
- **far out: look at this scan** – often a poor image
- **no fixed cut-off** – it differs between datasets
- **finds unusable scans** – Rosen 2018: AUC 0.98-0.99

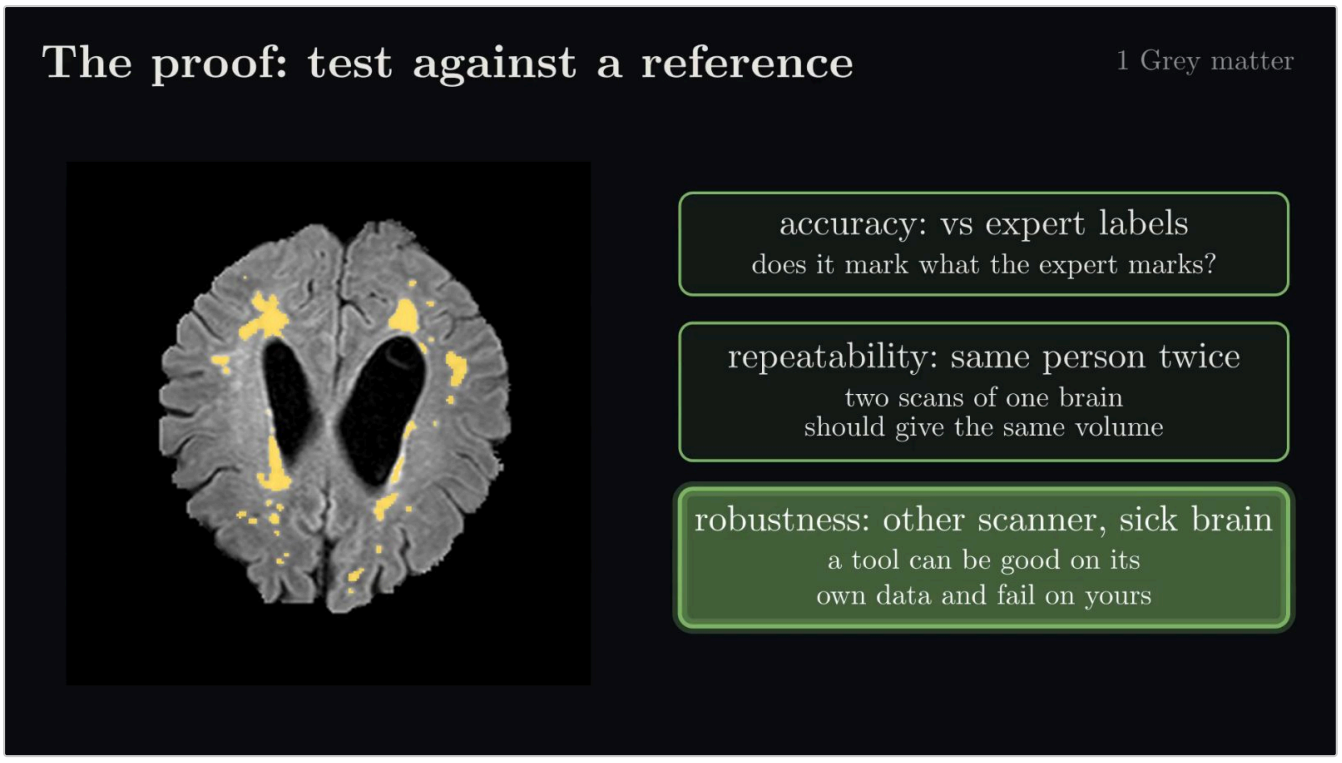
WHAT IS SAID

FreeSurfer has its own warning number: the Oiler number. Take a closed surface, like this cube. Count the corners, subtract the edges, add the faces. Eight minus twelve plus six is two. For a closed surface without holes, the answer is always two. Every handle lowers the number by two. FreeSurfer's first cortex surface is never perfect. It has faults, called defects. In this public stroke scan, the left side had twenty-six defects, and an Oiler number of minus thirty-eight. FreeSurfer repairs them, and the number is two again. One number alone says little. Compare it across your scans. A scan with far more defects than the others is often a poor image: look at it. There is no fixed cut-off. In a large study, this number found the unusable scans very well.

SOURCES

- Rosen et al. 2018, NeuroImage (Euler number)
  The Euler number of FreeSurfer's surface flags unusable scans very well (AUC 0.98-0.99).

16. The proof: test against a reference



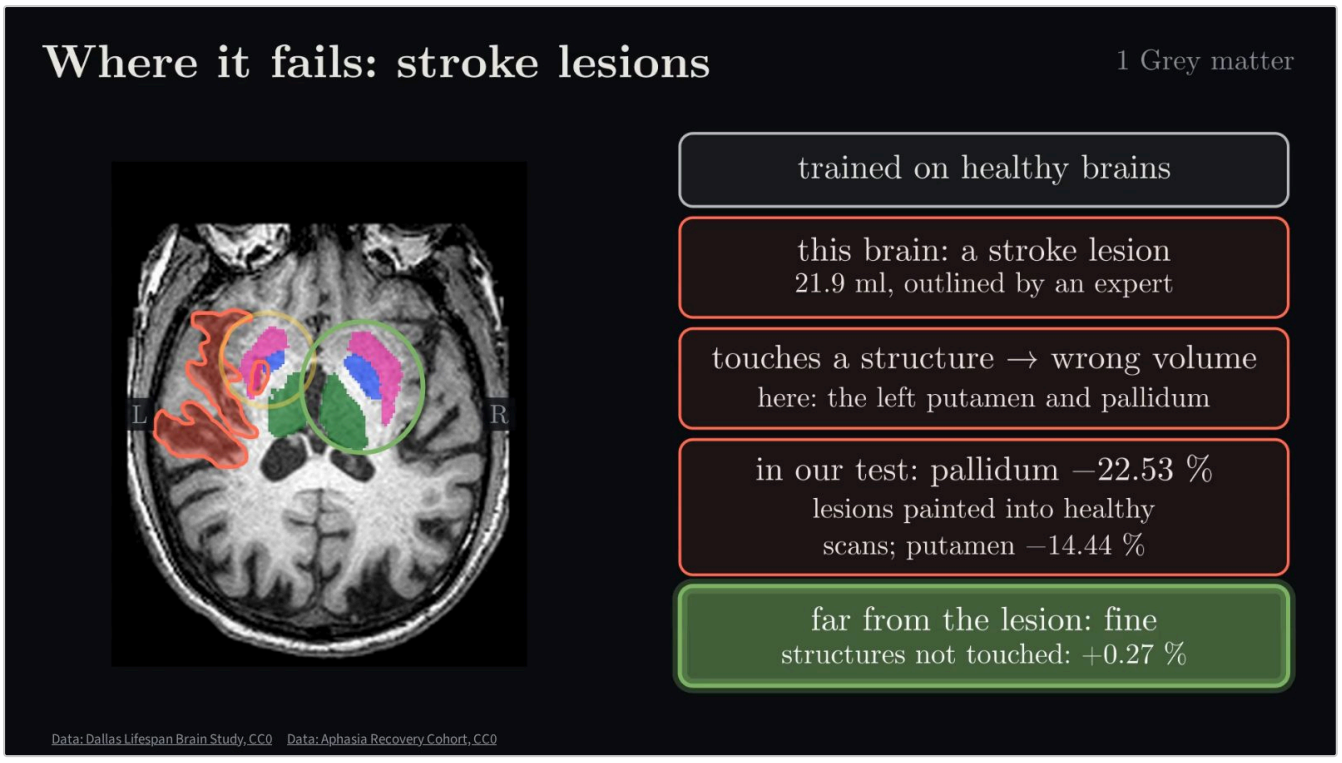
KEY POINTS

- **accuracy: vs expert labels** – does it mark what the expert marks?
- **repeatability: same person twice** – two scans of one brain should give the same volume
- **robustness: other scanner, sick brain** – a tool can be good on its own data and fail on yours

WHAT IS SAID

The strongest proof: compare with labels drawn by an expert. That is accuracy. Same person scanned twice: same result? That is repeatability. And does it still work on another scanner, or on a sick brain? Robustness.

17. Where it fails: stroke lesions



KEY POINTS

- **trained on healthy brains**
- **this brain: a stroke lesion**
- **lesion touches a structure → wrong volume**
- **pallidum −22.53 %**
- **far from the lesion: fine**

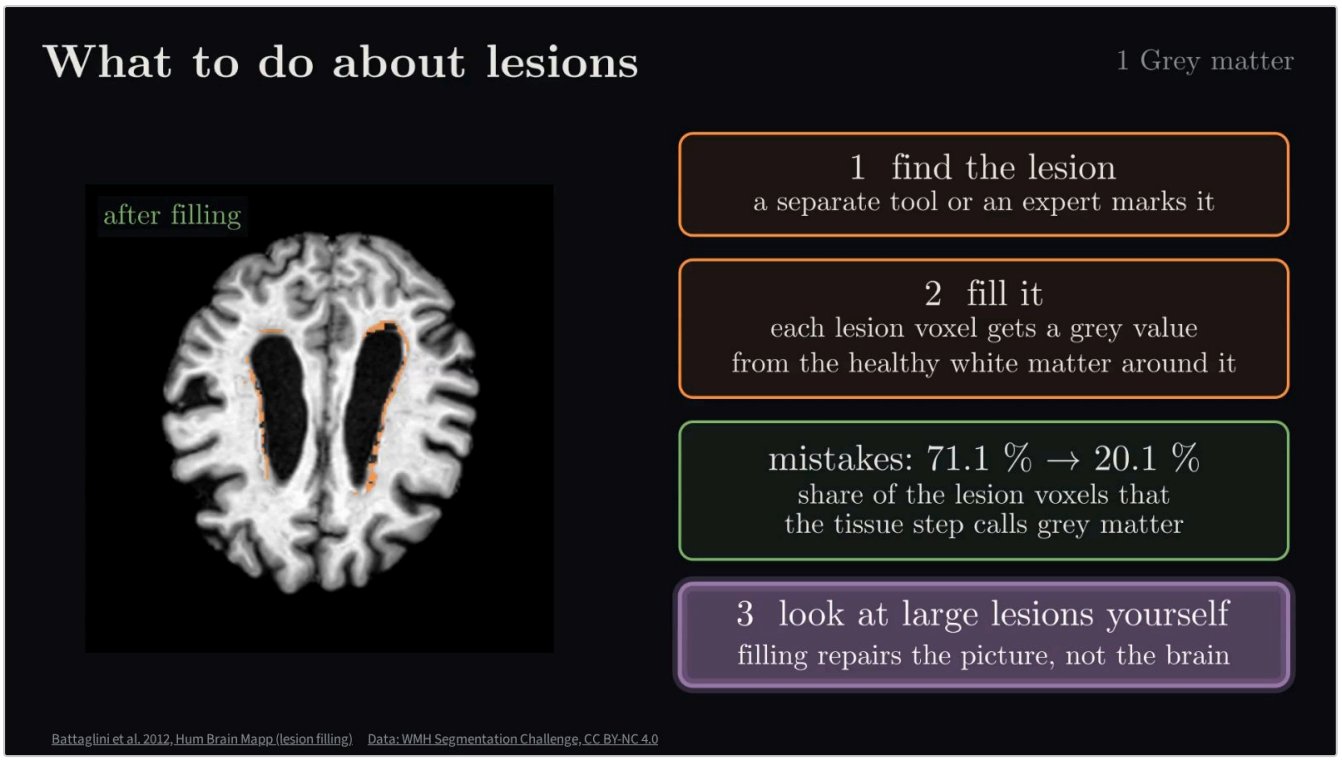
WHAT IS SAID

The tools learned from healthy brains. A large lesion is something they have never seen. This brain has a stroke lesion. Where the lesion touches a structure, the program gets its volume wrong. In our test, the pallidum came out about a fifth too small. Structures far from the lesion are fine.

SOURCES

- Data: Dallas Lifespan Brain Study, CC0
  Dallas Lifespan Brain Study: public MRI data (CC0), used for our grey-matter pictures.
- Data: Aphasia Recovery Cohort, CC0
  Aphasia Recovery Cohort: public stroke MRI with expert lesion masks (CC0).

18. What to do about lesions



KEY POINTS

- **1 find the lesion**
- **2 fill it with healthy white matter values**
- **grey matter mistakes: 71.1 % → 20.1 %**
- **3 look at large lesions yourself**

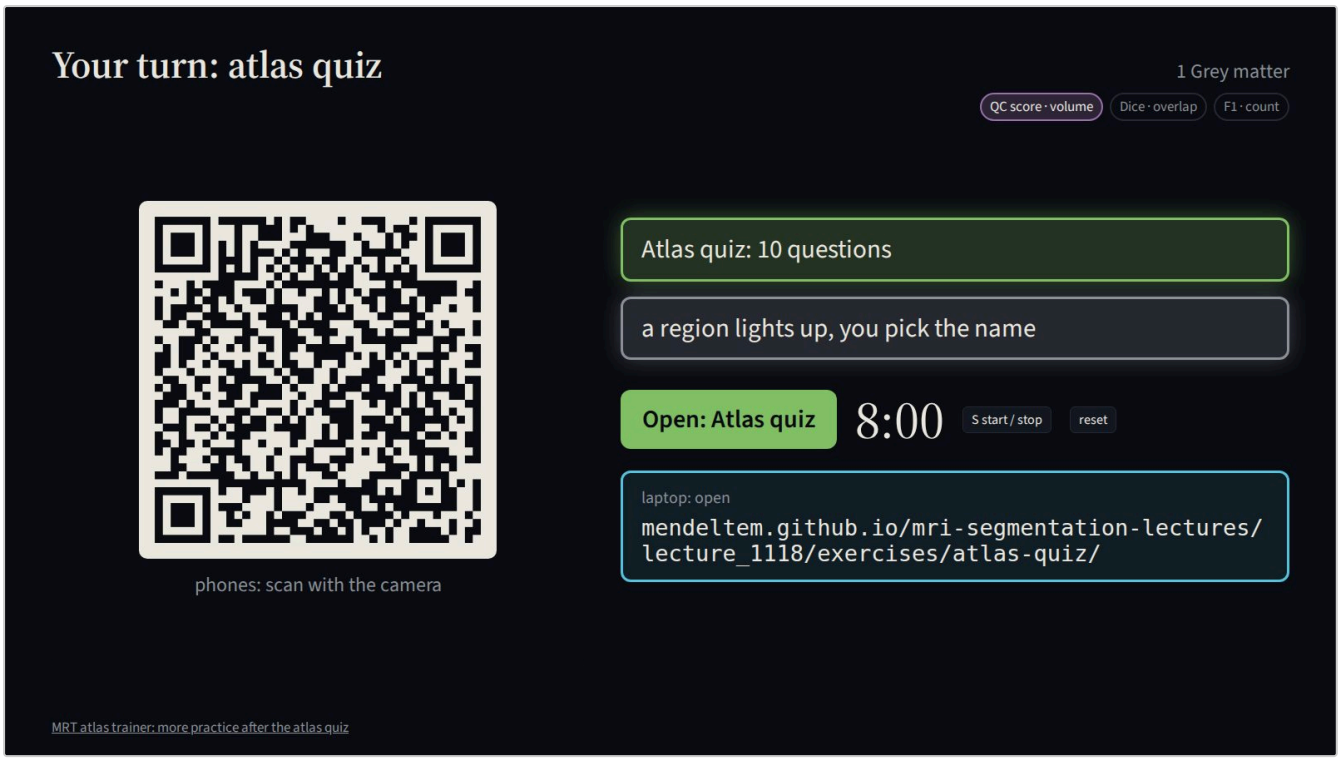
WHAT IS SAID

So: first find the lesion. An example with white matter lesions: on T one, they look grey. Then fill it. Zoom in. Each lesion voxel gets a new grey value, taken from the healthy white matter around it. The lesion disappears from the picture. Why? Before filling, the tissue step calls seventy one percent of the lesion voxels grey matter. After filling, twenty percent. And look at large lesions yourself.

SOURCES

- Battaglini et al. 2012, Hum Brain Mapp (lesion filling)
  Lesions bias tissue segmentation; filling them with normal-looking intensities removes most of the bias.
- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

19. Your turn: atlas quiz



> **Exercise: Atlas quiz** (8 min) – 10 questions: a region lights up, you pick its name.
> mendeltem.github.io/mri-segmentation-lectures/lecture_1118/exercises/atlas-quiz/

KEY POINTS

- **Atlas quiz: 10 questions**
- **a region lights up, you pick the name**

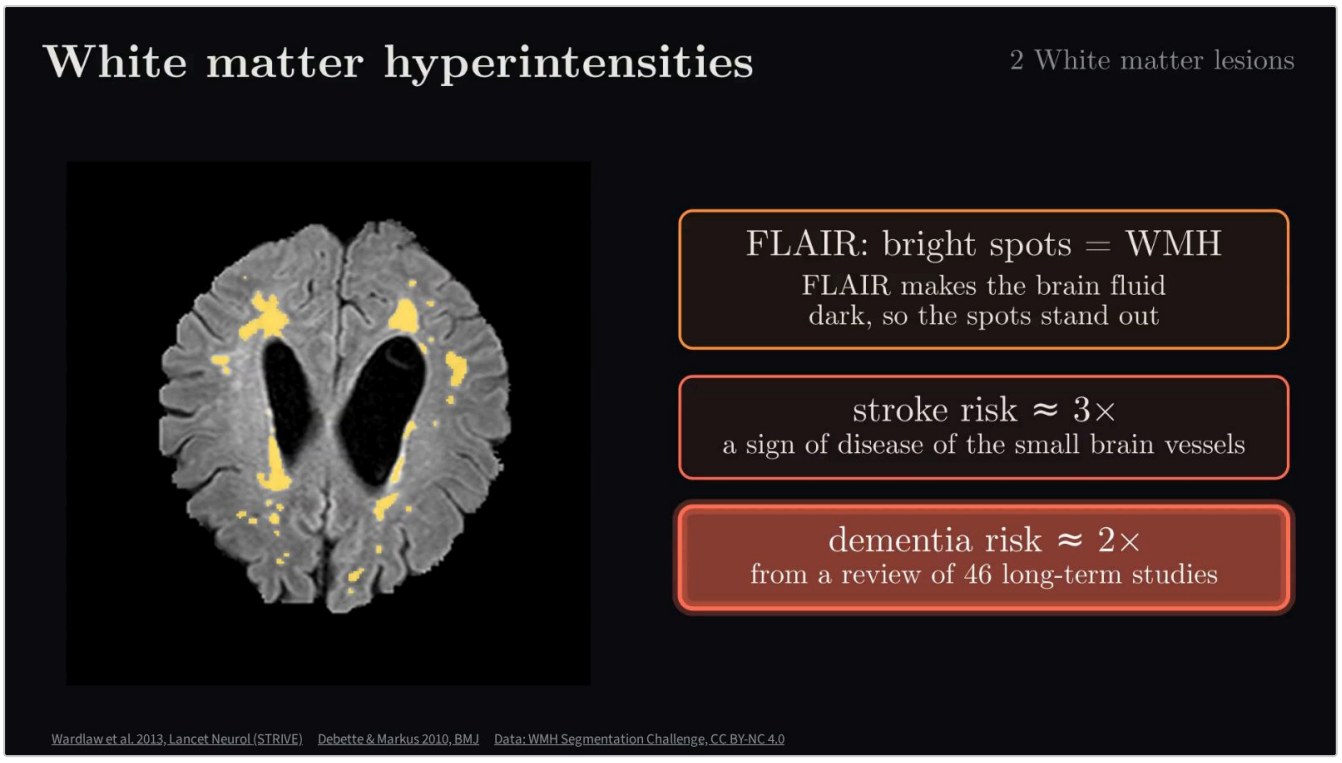
WHAT IS SAID

Now it is your turn. Open the atlas quiz on your phone or laptop. Ten questions: a region lights up, and you pick its name. You have eight minutes.

SOURCES

- MRT atlas trainer: more practice after the atlas quiz
  The full atlas trainer for home: explore, quiz and study modes with many more brain regions (interface in German).

20. White matter hyperintensities



KEY POINTS

- **FLAIR: bright spots = WMH** – FLAIR makes the brain fluid dark, so the spots stand out
- **stroke risk ≈ 3×** – a sign of disease of the small brain vessels
- **dementia risk ≈ 2×** – from a review of 46 long-term studies

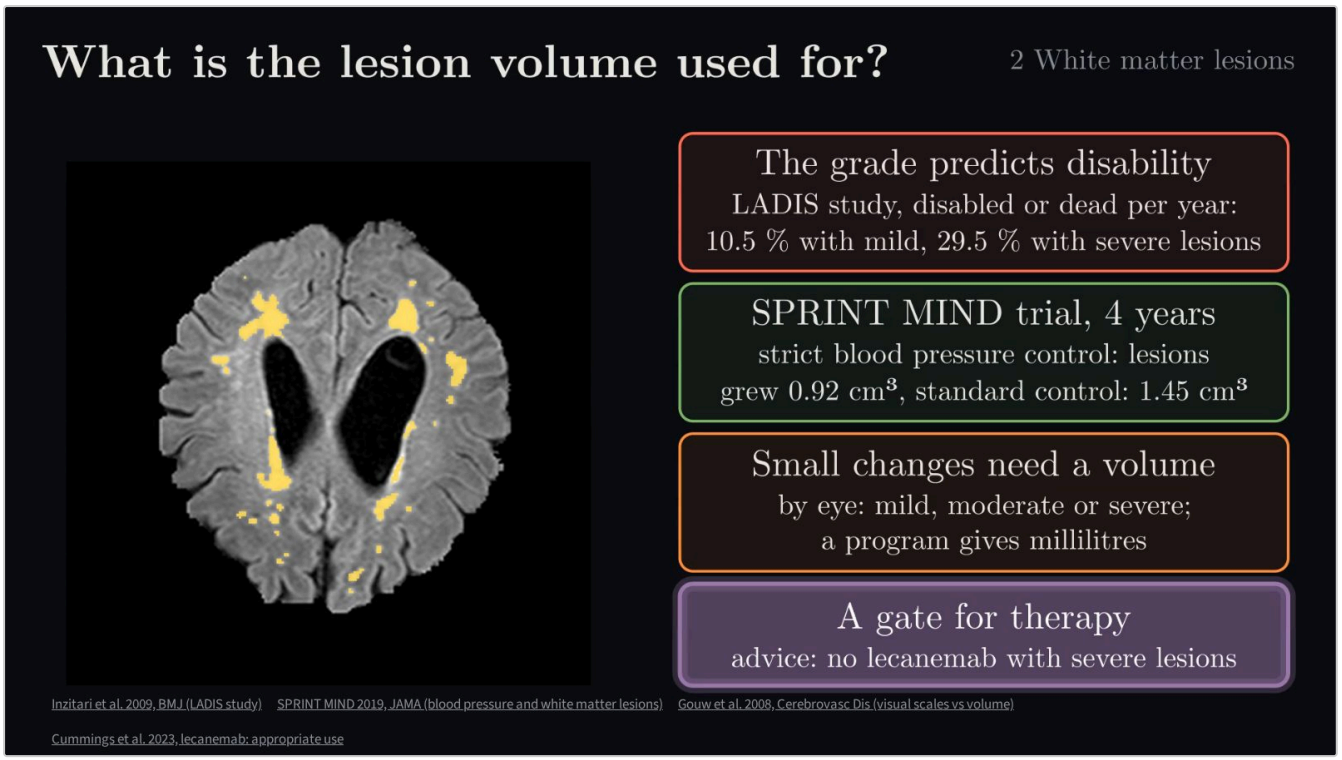
WHAT IS SAID

Part two. FLAIR is a scan that makes the brain fluid dark. So damaged white matter stands out as bright spots: white matter hyperintensities. They are a sign of small vessel disease. People with many of them have a higher risk of stroke and dementia.

SOURCES

- Wardlaw et al. 2013, Lancet Neurol (STRIVE)
  STRIVE consensus: names, definitions and imaging standards for small vessel disease (WMH, microbleeds).
- Debette & Markus 2010, BMJ
  Meta-analysis: white matter hyperintensities raise the risk of stroke (about 3x) and dementia (about 2x).
- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

21. What is the lesion volume used for?



KEY POINTS

- **The grade predicts disability** – LADIS study, disabled or dead per year: 10.5 % with mild, 29.5 % with severe lesions
- **SPRINT MIND trial, 4 years** – strict blood pressure control: lesions grew 0.92 cm³, standard control: 1.45 cm³
- **Small changes need a volume** – by eye: mild, moderate or severe; a program gives millilitres
- **A gate for therapy** – advice: no lecanemab with severe lesions

WHAT IS SAID

What do we do with this? In one study of older people, doctors graded the lesions by eye: mild, moderate or severe. With severe lesions, almost three times as many people lost their independence or died each year. Trials use the volume as an outcome: strict blood pressure control slowed the growth of the lesions. But the difference was small: about half a millilitre in four years. No grade by eye can show that. You need a measured volume. And for the new Alzheimer antibodies, severe lesions speak against the treatment.

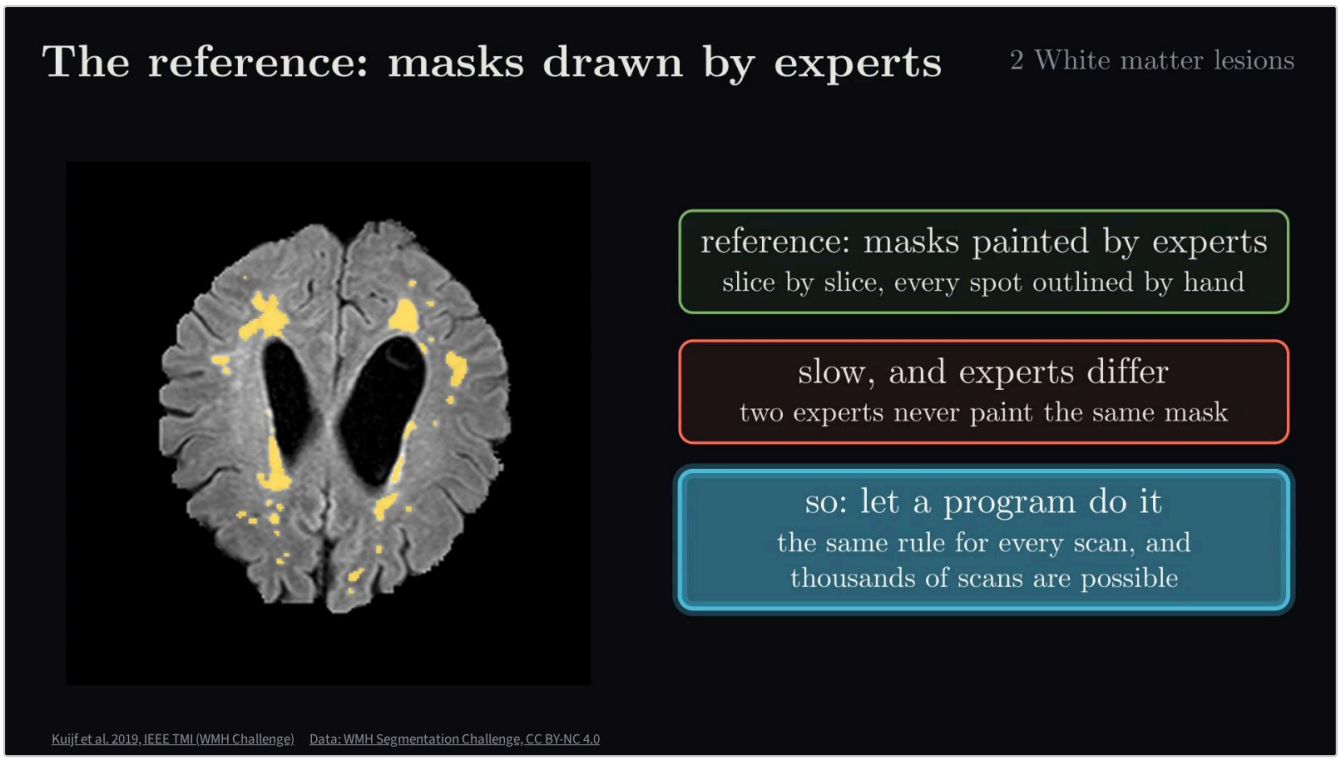
SOURCES

- Inzitari et al. 2009, BMJ (LADIS study)
  LADIS: older people living independently, 3 years follow-up; disability or death per year 10.5 % (mild), 15.1 % (moderate), 29.5 % (severe white matter changes).
- SPRINT MIND 2019, JAMA (blood pressure and white matter lesions)
  SPRINT MIND: intensive vs standard blood pressure control; lesion volume 4.57 to 5.49 cm3 vs 4.40 to 5.85 cm3 after about 4 years (difference -0.54 cm3, small).
- Gouw et al. 2008, Cerebrovasc Dis (visual scales vs volume)

Measuring the change of lesion volume was reliable (ICC 0.88); judging progression with visual scales was less reliable (20 scan pairs).

- Cummings et al. 2023, lecanemab: appropriate use
  Appropriate use recommendations for lecanemab: no treatment with more than 4 microhaemorrhages.

22. The reference: masks drawn by experts



KEY POINTS

- **reference: masks painted by experts** – slice by slice, every spot outlined by hand
- **slow, and experts differ** – two experts never paint the same mask
- **so: let a program do it** – the same rule for every scan, and thousands of scans are possible

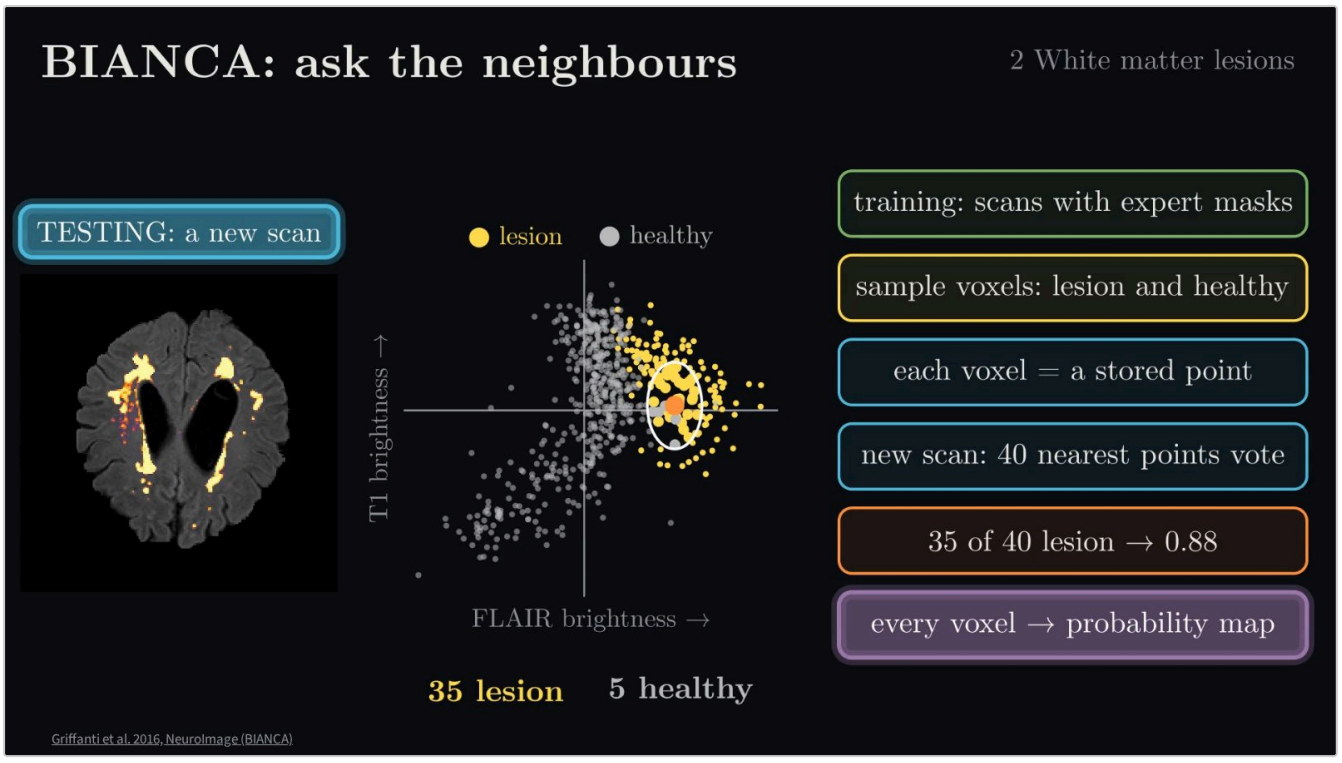
WHAT IS SAID

The reference is a mask painted by an expert, by hand, slice by slice. That is slow, and two experts never paint exactly the same. So we want a program to do it.

SOURCES

- Kuijf et al. 2019, IEEE TMI (WMH Challenge)
  WMH Segmentation Challenge: 60 training and 110 test scans, 5 scanners; best method Dice 0.80, second experts 0.77-0.79.
- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

23. BIANCA: ask the neighbours



KEY POINTS

- **training: scans with expert masks**
- **sample voxels: lesion and healthy**
- **each voxel = a point**
- **new scan: 40 nearest points vote**
- **35 of 40 lesion → 0.88**
- **every voxel → probability map**

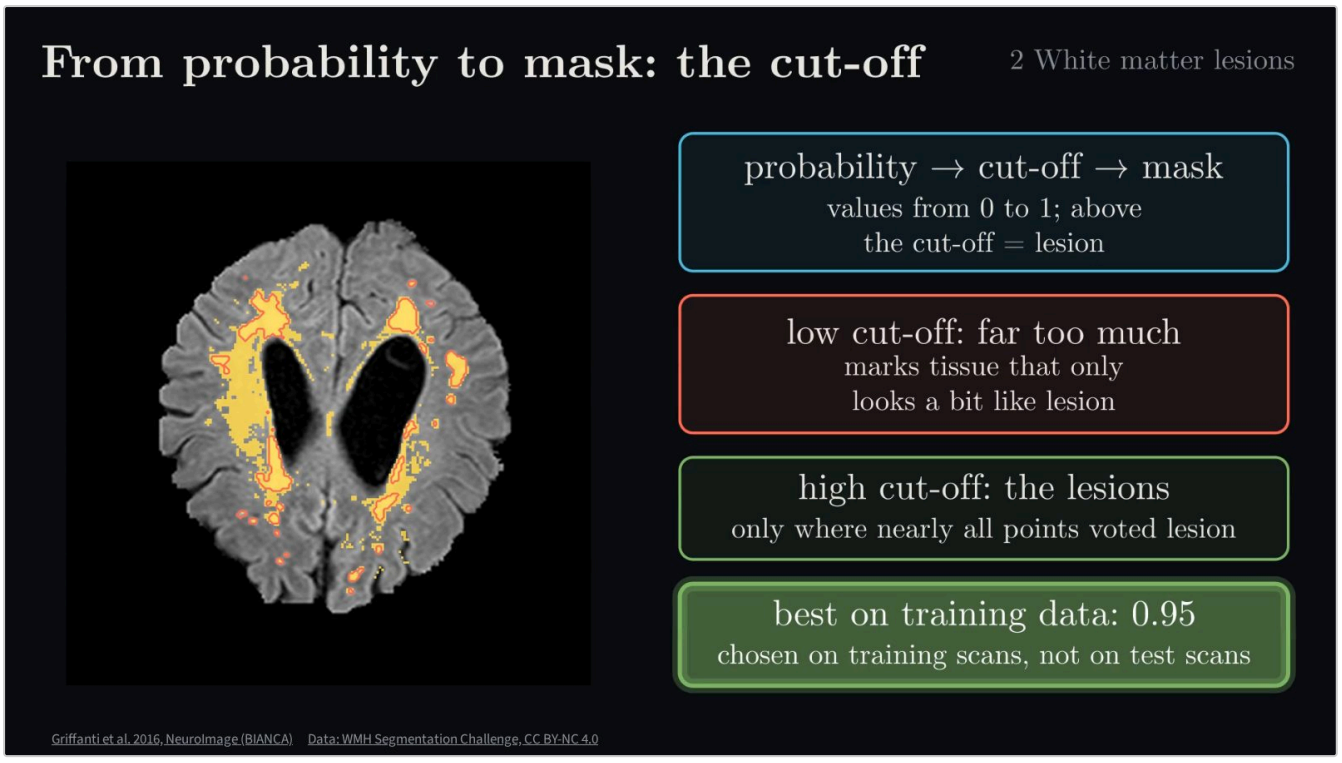
WHAT IS SAID

BIANCA learns from examples. In training scans, an expert has marked the lesions. BIANCA takes sample voxels, lesion and healthy, and stores each one as a point: its brightness and its position. Now a new scan, without a mask. For one voxel, the forty nearest stored points vote. Thirty five say lesion: probability zero point eight eight. BIANCA does this for every voxel.

SOURCES

- Griffanti et al. 2016, NeuroImage (BIANCA)
  BIANCA: k-nearest-neighbour WMH segmentation in FSL; Dice 0.76 in one cohort, 0.52 in another.

24. From probability to mask: the cut-off



KEY POINTS

- **probability → cut-off → mask** – values from 0 to 1; above the cut-off = lesion
- **low cut-off: far too much** – marks tissue that only looks a bit like lesion
- **high cut-off: the lesions** – only where nearly all points voted lesion
- **best on training data: 0.95** – chosen on training scans, not on test scans

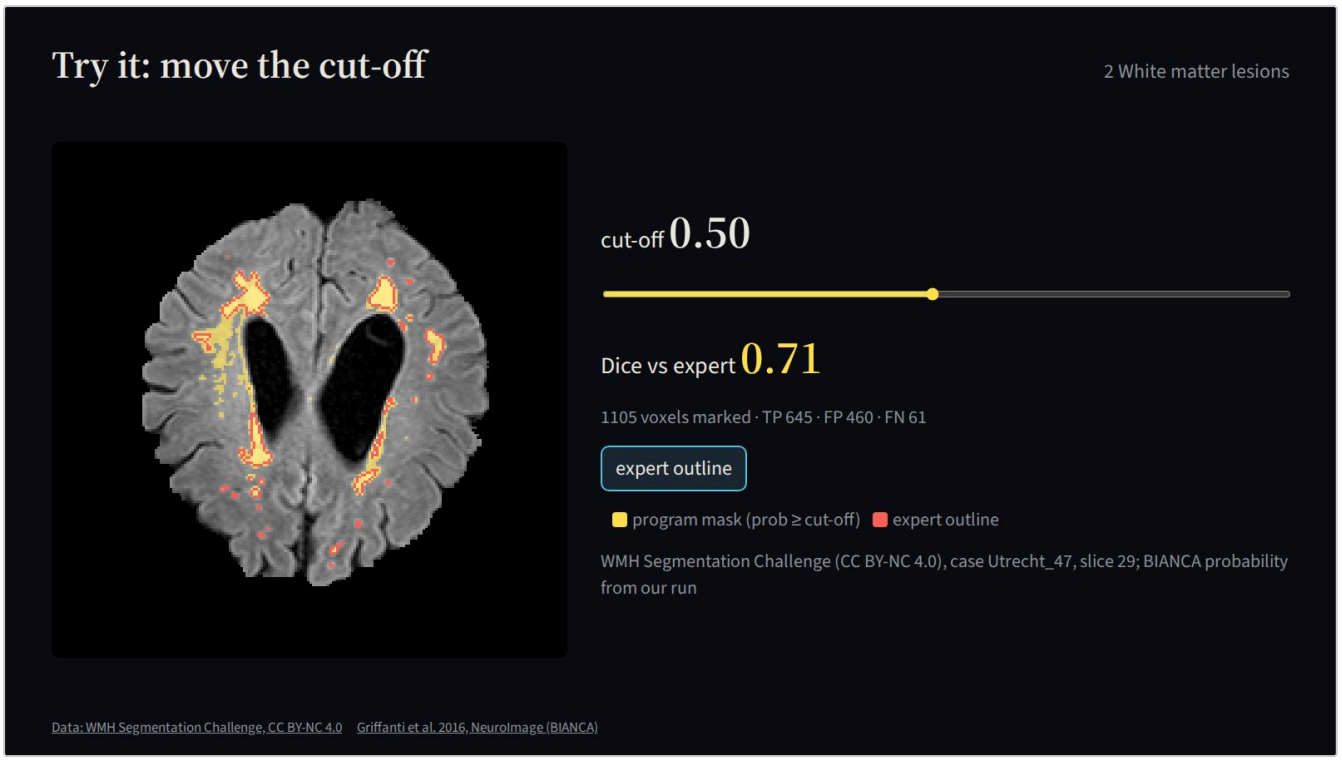
WHAT IS SAID

This gives a probability map: every voxel has a value between zero and one. A cut off turns it into a mask: above the cut off counts as lesion. A low cut off also marks tissue that only looks a little like a lesion. A high one keeps the clear lesions. Zero point nine five was best, chosen on the training scans.

SOURCES

- Griffanti et al. 2016, NeuroImage (BIANCA)
  BIANCA: k-nearest-neighbour WMH segmentation in FSL; Dice 0.76 in one cohort, 0.52 in another.
- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

25. Try it: move the cut-off



> Interactive page: try it in the online deck – mendeltem.github.io/mri-segmentation-lectures/lecture_1118/deck/#try_cutoff

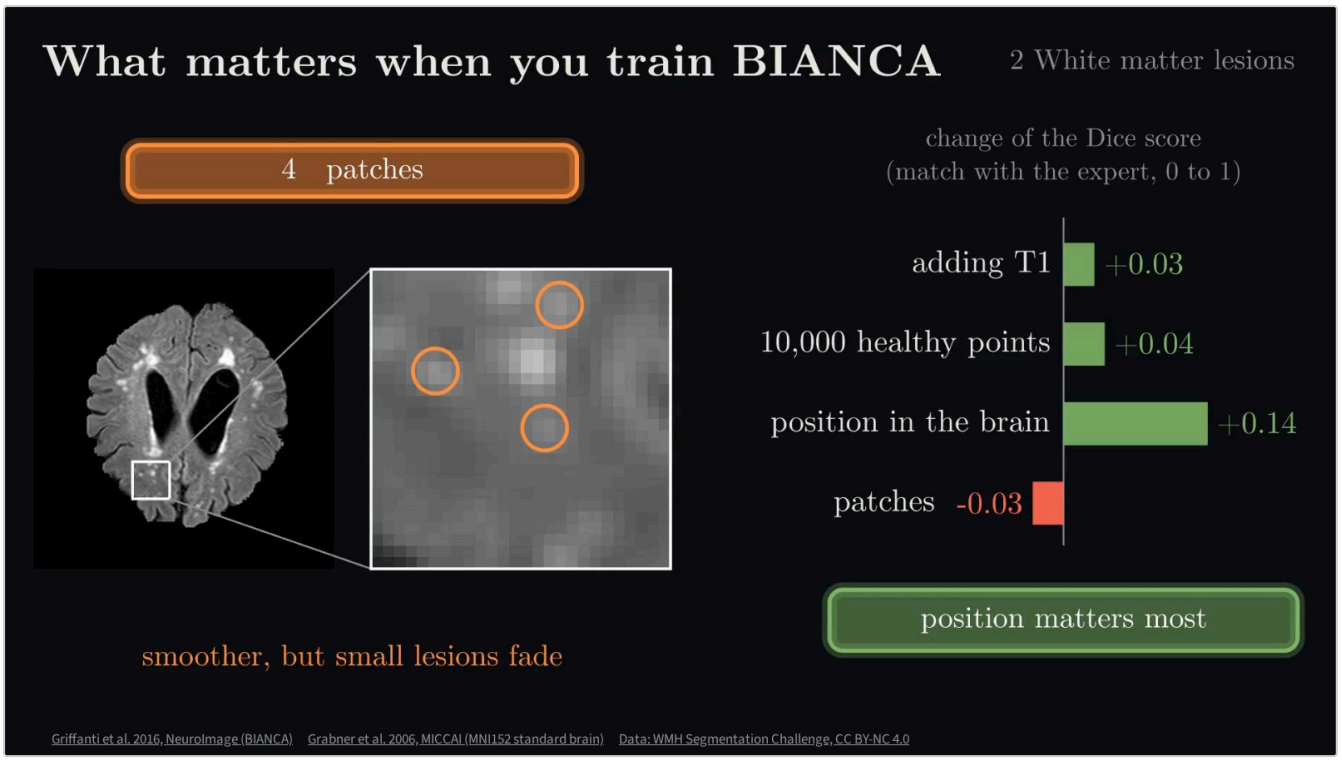
WHAT IS SAID

Move the slider. Low cut-off: the mask floods the white matter and Dice drops. High cut-off: only the lesions stay. Find the best Dice on this slice; BIANCA's chosen cut-off was 0.95.

SOURCES

- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.
- Griffanti et al. 2016, NeuroImage (BIANCA)
  BIANCA: k-nearest-neighbour WMH segmentation in FSL; Dice 0.76 in one cohort, 0.52 in another.

26. What matters when you train BIANCA



KEY POINTS

- **adding T1: +0.03**
- **10,000 healthy points: +0.04**
- **position in the brain: +0.14**
- **patches: −0.03**

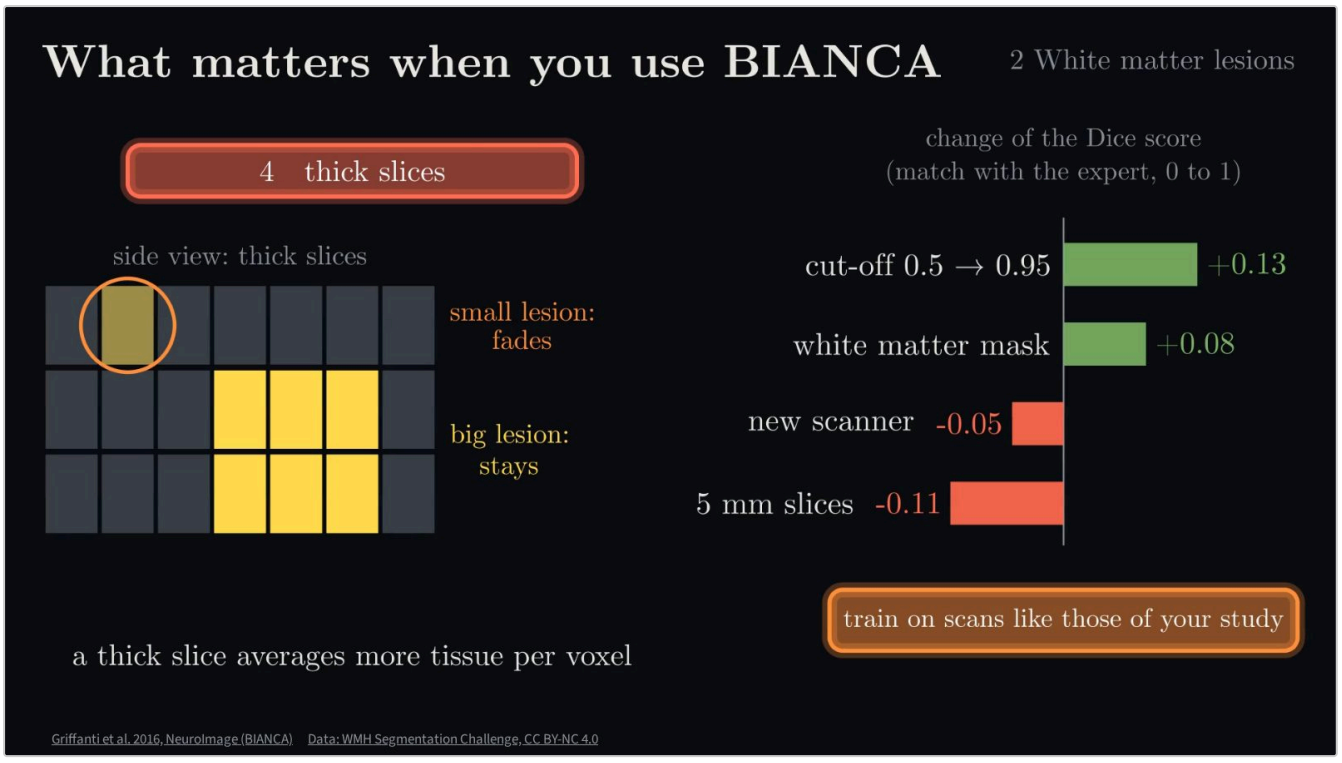
WHAT IS SAID

BIANCA has settings. We tried them, and each time measured how well the result matches the expert. First: which images? With FLAIR only, each voxel is one number. Adding the T one gives it a second number. That helped a little. Second: how many healthy examples? Healthy tissue looks different from place to place. With ten thousand examples per scan instead of two thousand, the result got a little better. Third: position. Lesions have typical places. This map adds up the lesions of all training scans. Most lie around the ventricles. When BIANCA also knows where a voxel lies, the score rises the most. Last: patches. BIANCA can also look at the three by three neighbourhood of a voxel, and use its average. That smooths noise, but small lesions fade. In our runs it made the result slightly worse.

SOURCES

- Griffanti et al. 2016, NeuroImage (BIANCA)
  BIANCA: k-nearest-neighbour WMH segmentation in FSL; Dice 0.76 in one cohort, 0.52 in another.
- Grabner et al. 2006, MICCAI (MNI152 standard brain)
  The MNI152 standard brain (average of 152 healthy brains): scans are aligned to it, so every voxel gets a position that means the same place in every person.
- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

27. What matters when you use BIANCA



KEY POINTS

- **cut-off 0.5 → 0.95: +0.13**
- **white matter mask: +0.08**
- **new scanner: −0.05**
- **5 mm slices: −0.11**

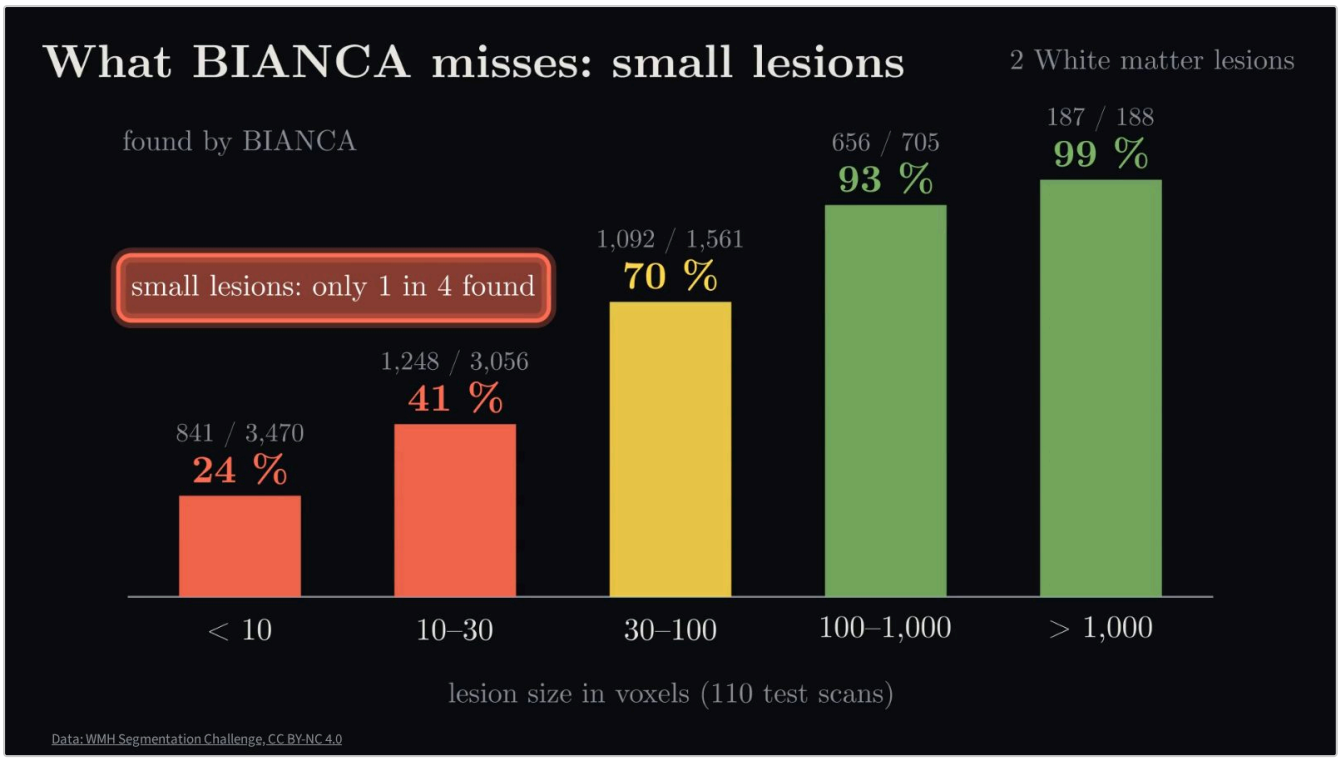
WHAT IS SAID

After training, BIANCA gives a probability for every voxel. The cut off decides what counts as lesion. At zero point five, too much is marked. A high cut off, zero point nine five, was much better. Some marks are still wrong: they lie outside the white matter, for example in the cortex. A white matter mask removes them. The true lesions stay. Now your data. BIANCA learned on scans from certain scanners. A new scanner gives another brightness and another contrast. The stored examples fit less well, and the score drops. And slice thickness. A thick slice averages more tissue in every voxel. Big lesions stay. Small lesions fade away. That cost the most. So train BIANCA on scans like the ones in your study.

SOURCES

- Griffanti et al. 2016, NeuroImage (BIANCA)
  BIANCA: k-nearest-neighbour WMH segmentation in FSL; Dice 0.76 in one cohort, 0.52 in another.
- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

28. What BIANCA misses: small lesions



KEY POINTS

- **over 1,000 voxels: 99 % found**
- **under 10 voxels: 24 % found**

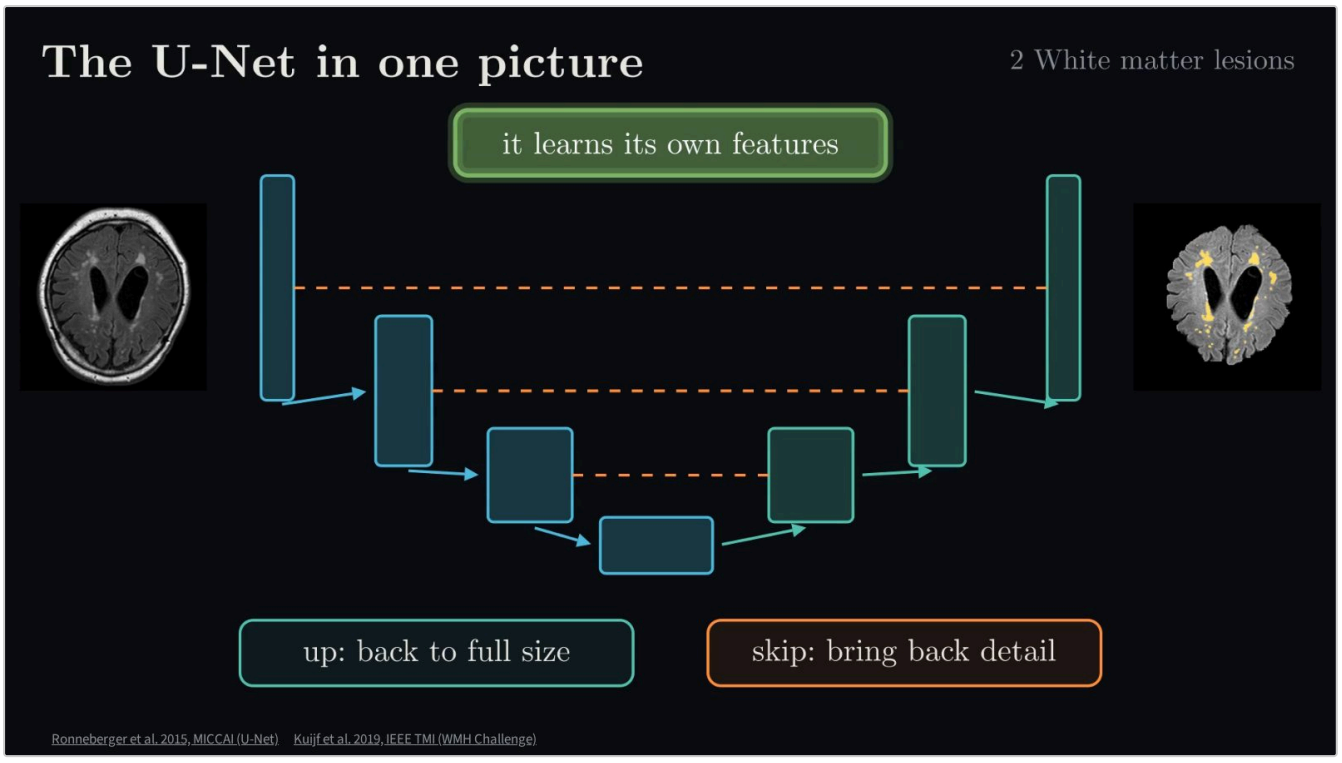
WHAT IS SAID

Which lesions does BIANCA find? The big ones, almost always. But of the smallest lesions, under ten voxels, only one in four. That is why small lesions pull Dice down.

SOURCES

- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

29. The U-Net in one picture



KEY POINTS

- **down: shrink, collect context**
- **up: back to full size**
- **skip connections: fine detail**
- **learns its own features**

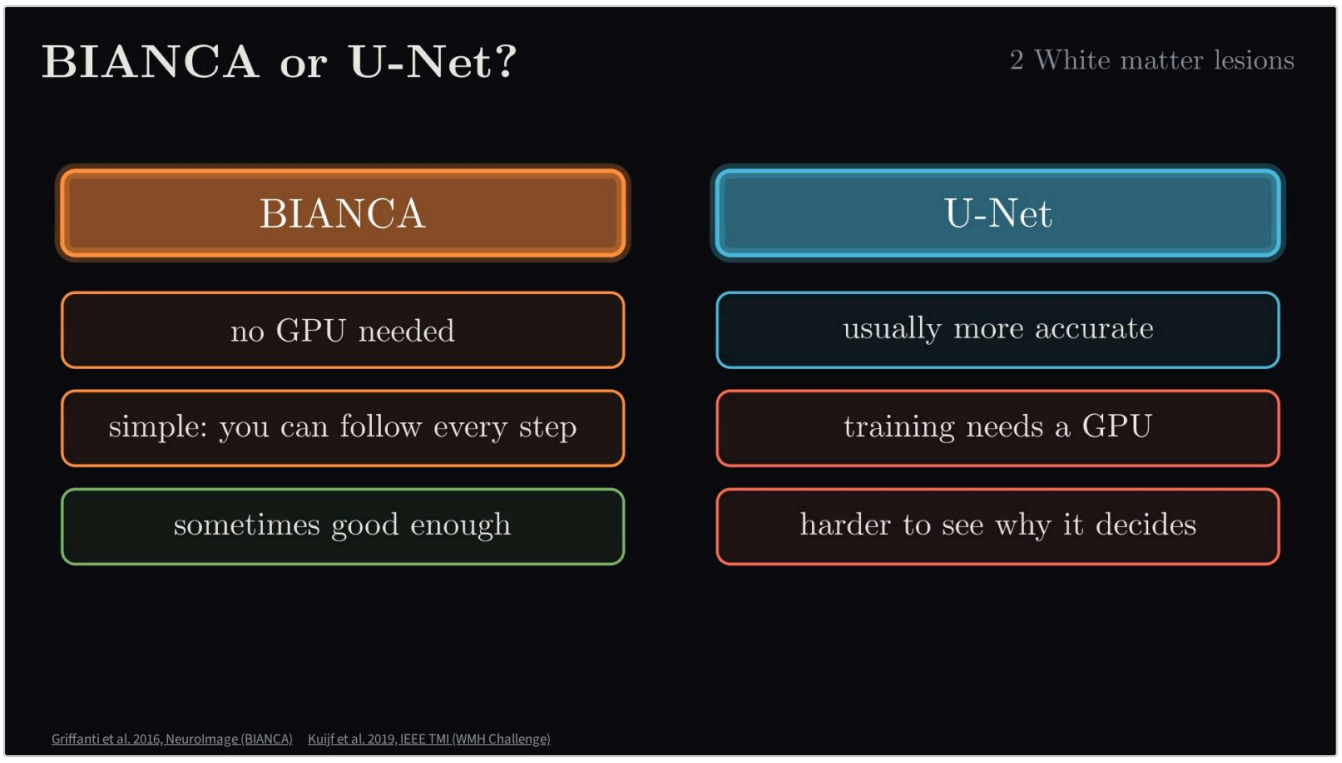
WHAT IS SAID

A U-Net goes down, shrinking the image to see context. Then it goes up again, and skip connections bring back the detail. It learns its own features. Today, the best lesion tools work like this.

SOURCES

- Ronneberger et al. 2015, MICCAI (U-Net)
  U-Net: the encoder-decoder network with skip connections for biomedical image segmentation.
- Kuijf et al. 2019, IEEE TMI (WMH Challenge)
  WMH Segmentation Challenge: 60 training and 110 test scans, 5 scanners; best method Dice 0.80, second experts 0.77-0.79.

30. BIANCA or U-Net?



KEY POINTS

- **BIANCA: no GPU needed**
- **simple: you can follow every step**
- **sometimes good enough**
- **U-Net: usually more accurate**
- **training needs a GPU**
- **harder to see why it decides**

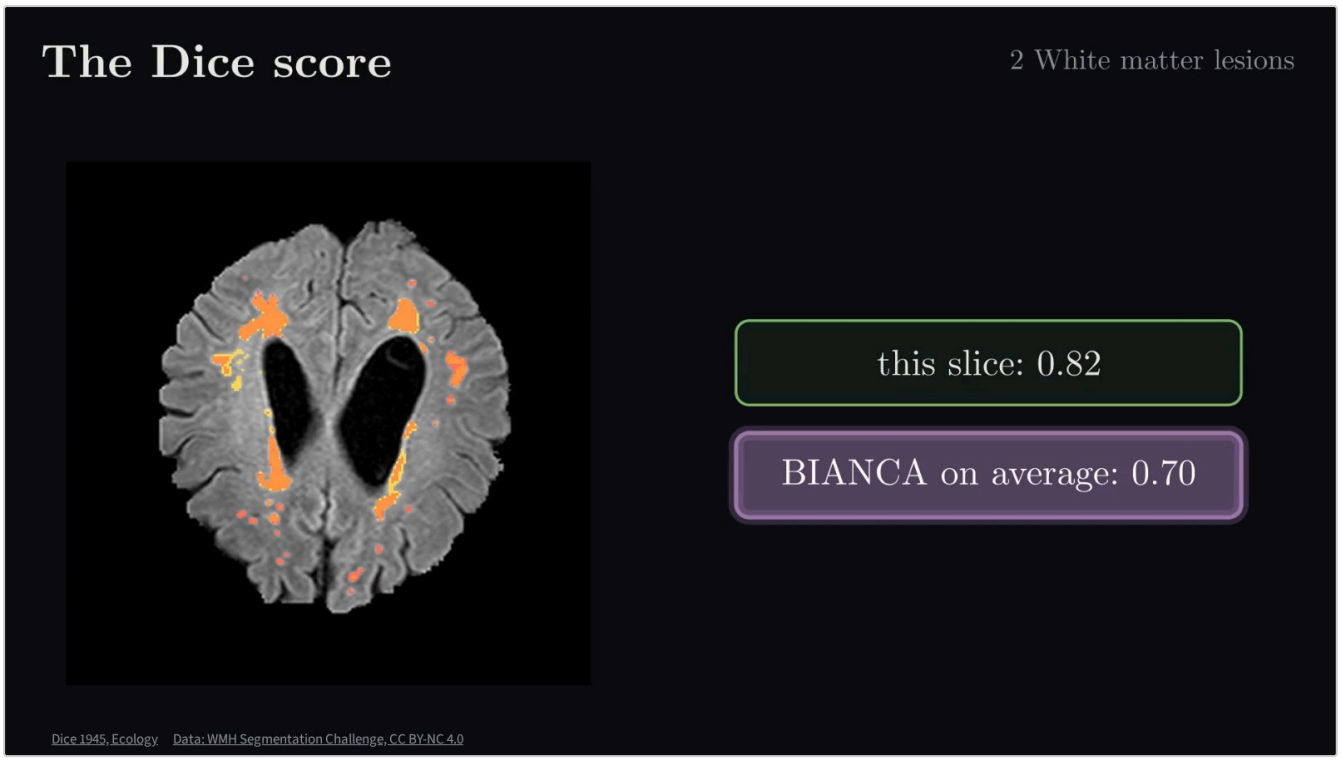
WHAT IS SAID

So, BIANCA or U-Net? BIANCA needs no graphics card. It is simple: you can follow every step. And sometimes, it is good enough. A U-Net is usually more accurate. But training it needs a graphics card, and it is harder to see why it decides.

SOURCES

- Griffanti et al. 2016, NeuroImage (BIANCA)
  BIANCA: k-nearest-neighbour WMH segmentation in FSL; Dice 0.76 in one cohort, 0.52 in another.
- Kuijf et al. 2019, IEEE TMI (WMH Challenge)
  WMH Segmentation Challenge: 60 training and 110 test scans, 5 scanners; best method Dice 0.80, second experts 0.77-0.79.

31. The Dice score



KEY POINTS

- **TP: in both**
- **FP: only the program**
- **FN: missed**
- **Dice = 2·TP / (2·TP + FP + FN)**
- **this slice: 0.82**
- **BIANCA on average: 0.70**

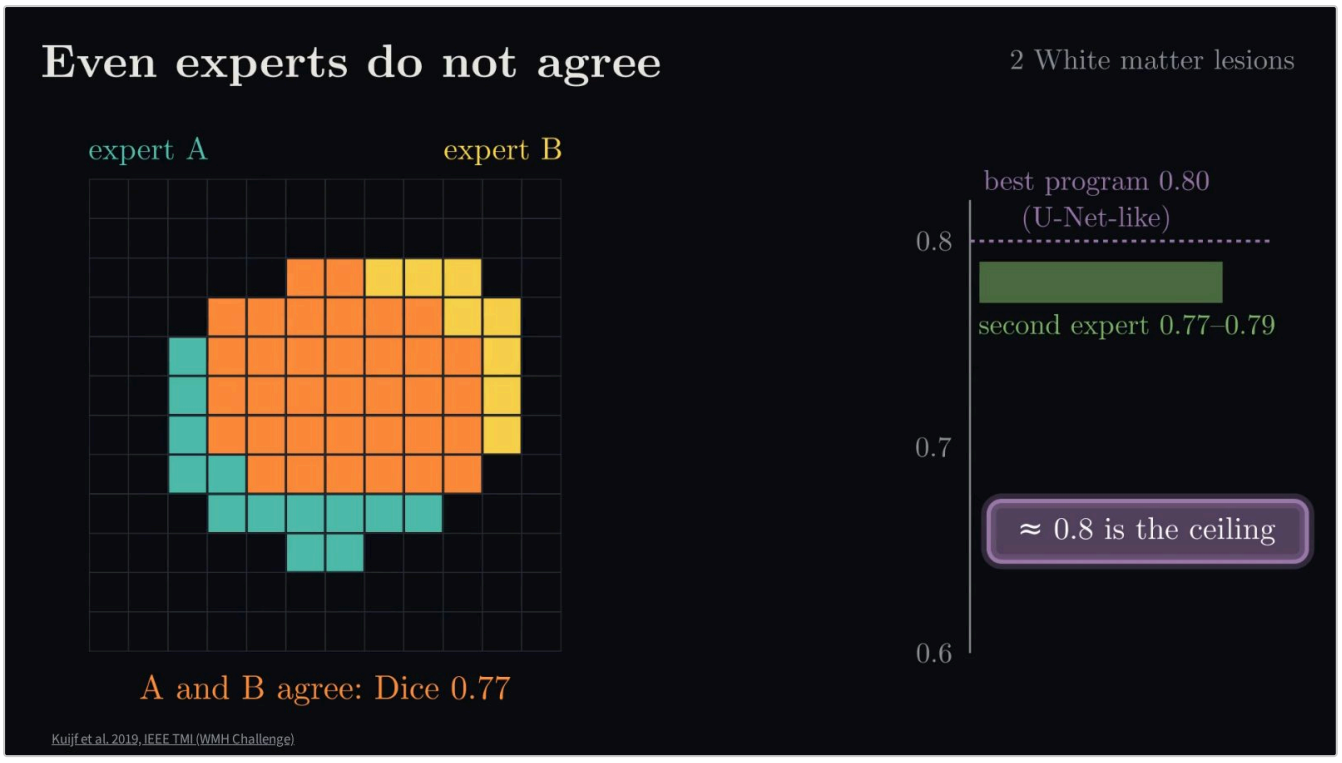
WHAT IS SAID

Put the mask on the reference: hits in both, extra marks, and misses. Dice is two times the hits, divided by both mask sizes together. One is perfect. This slice: zero point eight two. BIANCA on average: zero point seven.

SOURCES

- Dice 1945, Ecology
  Dice's 1945 overlap coefficient, first used to measure how often two species occur together.
- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

32. Even experts do not agree



KEY POINTS

- **second expert vs reference: 0.77–0.79**
- **best program (U-Net-like): 0.80**

WHAT IS SAID

Even experts do not agree. A second expert, on the same scans, reached only zero point seven seven to zero point seven nine against the reference. So about zero point eight is the ceiling. The best program, a U-Net, reached it.

SOURCES

- Kuijf et al. 2019, IEEE TMI (WMH Challenge)
  WMH Segmentation Challenge: 60 training and 110 test scans, 5 scanners; best method Dice 0.80, second experts 0.77-0.79.

33. Dice is hard on small lesions



KEY POINTS

- **10 × 10 voxels, shifted by 1**
- **Dice 0.90** – 90 of 100 voxels still overlap
- **2 × 2 voxels, same shift**
- **Dice 0.50** – only 2 of 4 voxels still overlap
- **small lesions are hard on Dice** – the same small error costs a small lesion much more

WHAT IS SAID

Shift a big lesion by one voxel. Ninety of a hundred voxels still overlap: Dice is zero point nine. The same shift on a tiny lesion. Only two of four voxels still overlap: Dice is zero point five. Small lesions are hard on Dice.

34. Try it: shift the mask



> Interactive page: try it in the online deck – mendeltem.github.io/mri-segmentation-lectures/lecture_1118/deck/#try_dice

WHAT IS SAID

Shift the program's mask with the buttons. One voxel costs the big lesion little and the small lesion a lot. Two voxels and the small lesion has no overlap at all.

35. Your turn: segment the lesions



> **Exercise: Lesion exercise** (6 min) – 1 patient, 3 slices. Short tutorial, paint, get your Dice.
> mendeltem.github.io/mri-segmentation-lectures/lecture_1118/exercises/wmh-dice/

KEY POINTS

- **Lesion exercise: 1 patient, 3 slices**
- **tutorial → paint → your Dice**

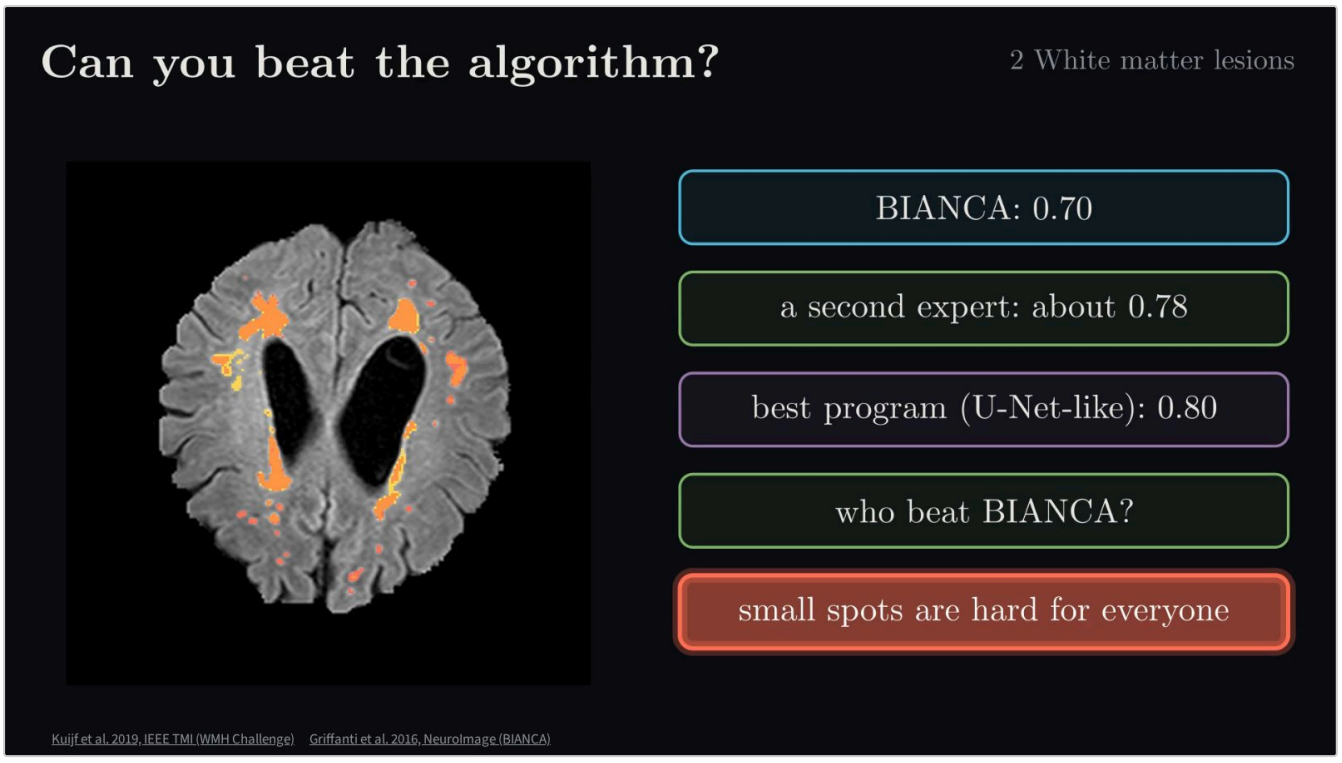
WHAT IS SAID

Your turn. Open the lesion exercise. A short tutorial first, then paint the lesions on three slices of one patient, and get your Dice. You have six minutes.

SOURCES

- Data: WMH Segmentation Challenge, CC BY-NC 4.0
  The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.

36. Can you beat the algorithm?



KEY POINTS

- **BIANCA: 0.70**
- **a second expert: about 0.78**
- **best program (U-Net-like): 0.80**
- **who beat BIANCA?**
- **small spots are hard for everyone**

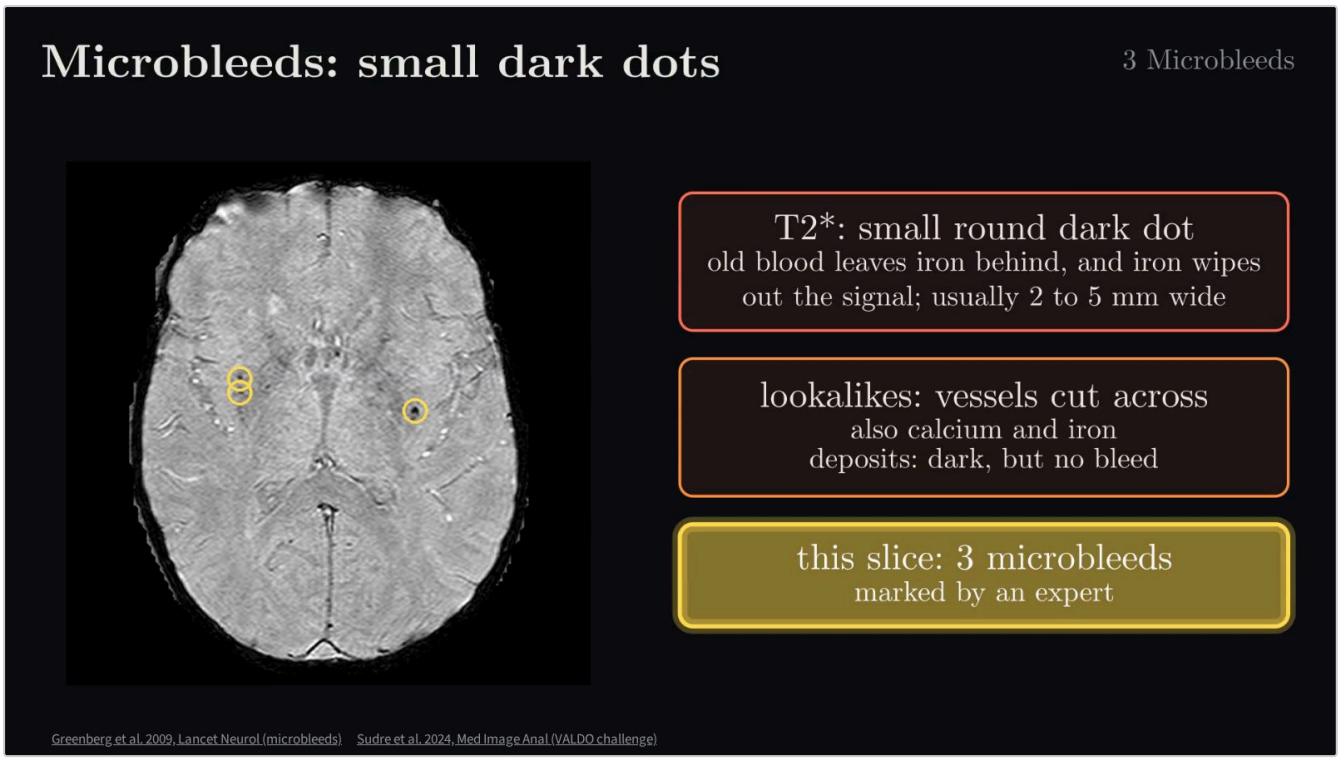
WHAT IS SAID

How did you do? BIANCA got zero point seven, a second expert about zero point seven eight, the best program, a U-Net, zero point eight. Hands up, who beat BIANCA? The small spots are hard for everyone.

SOURCES

- Kuijf et al. 2019, IEEE TMI (WMH Challenge)
  WMH Segmentation Challenge: 60 training and 110 test scans, 5 scanners; best method Dice 0.80, second experts 0.77-0.79.
- Griffanti et al. 2016, NeuroImage (BIANCA)
  BIANCA: k-nearest-neighbour WMH segmentation in FSL; Dice 0.76 in one cohort, 0.52 in another.

37. Microbleeds: small dark dots



KEY POINTS

- **T2*: small round dark dot** – old blood leaves iron behind, and iron wipes out the signal; usually 2 to 5 mm wide
- **lookalikes: vessels cut across** – also calcium and iron deposits: dark, but no bleed
- **this slice: 3 microbleeds** – marked by an expert

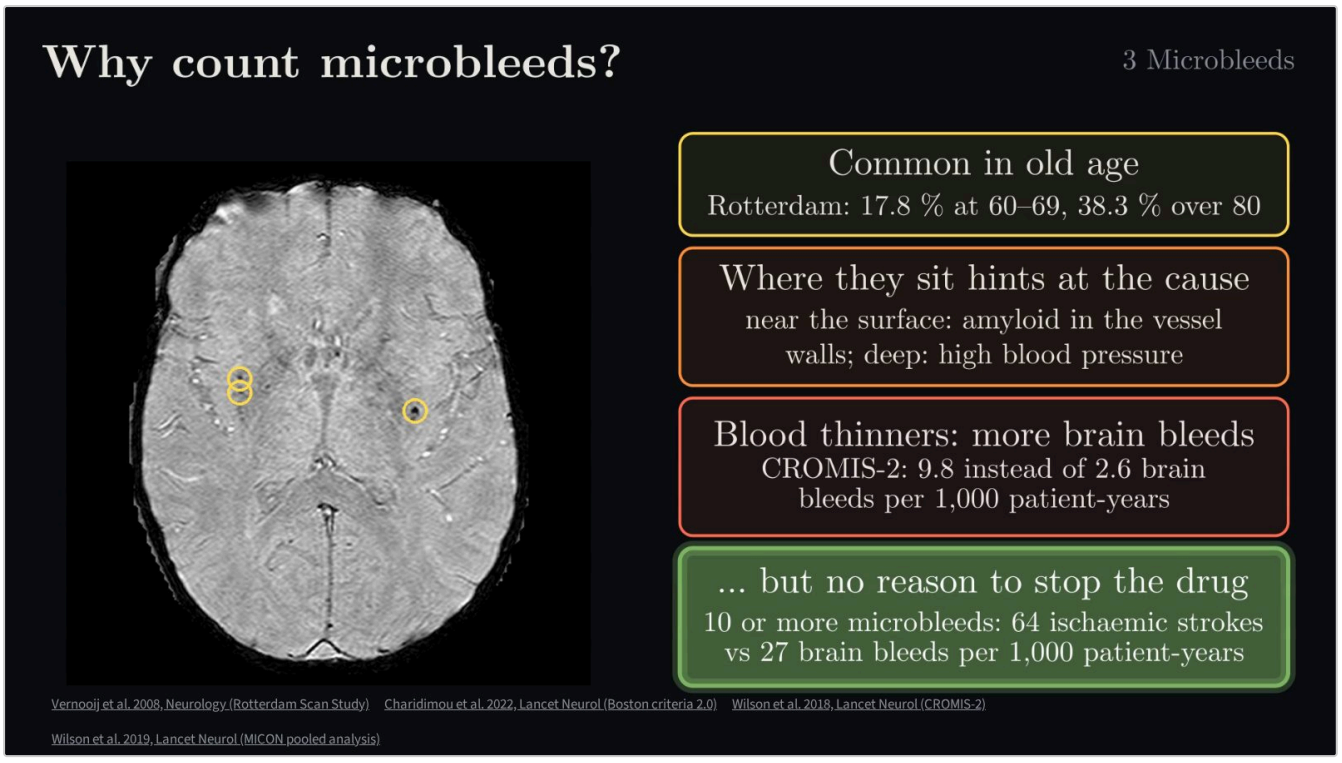
WHAT IS SAID

Part three. A microbleed is a tiny old bleed. It leaves iron behind, and iron disturbs the signal. So on T two star it is a small, round, dark dot, usually two to five millimetres wide. Watch for lookalikes, like vessels cut across. This slice has three.

SOURCES

- Greenberg et al. 2009, Lancet Neurol (microbleeds)
  Review of cerebral microbleeds: definition, detection on T2*/SWI, mimics and clinical meaning.
- Sudre et al. 2024, Med Image Anal (VALDO challenge)
  VALDO challenge (MICCAI 2021): automatic detection of small vessel lesions, including microbleeds.

38. Why count microbleeds?



KEY POINTS

- **Common in old age** – Rotterdam: 17.8 % at 60–69, 38.3 % over 80
- **Where they sit hints at the cause** – near the surface: amyloid in the vessel walls; deep: high blood pressure
- **Blood thinners: more brain bleeds** – CROMIS-2: 9.8 instead of 2.6 brain bleeds per 1,000 patient-years
- **… but no reason to stop the drug** – 10 or more microbleeds: 64 ischaemic strokes vs 27 brain bleeds per 1,000 patient-years

WHAT IS SAID

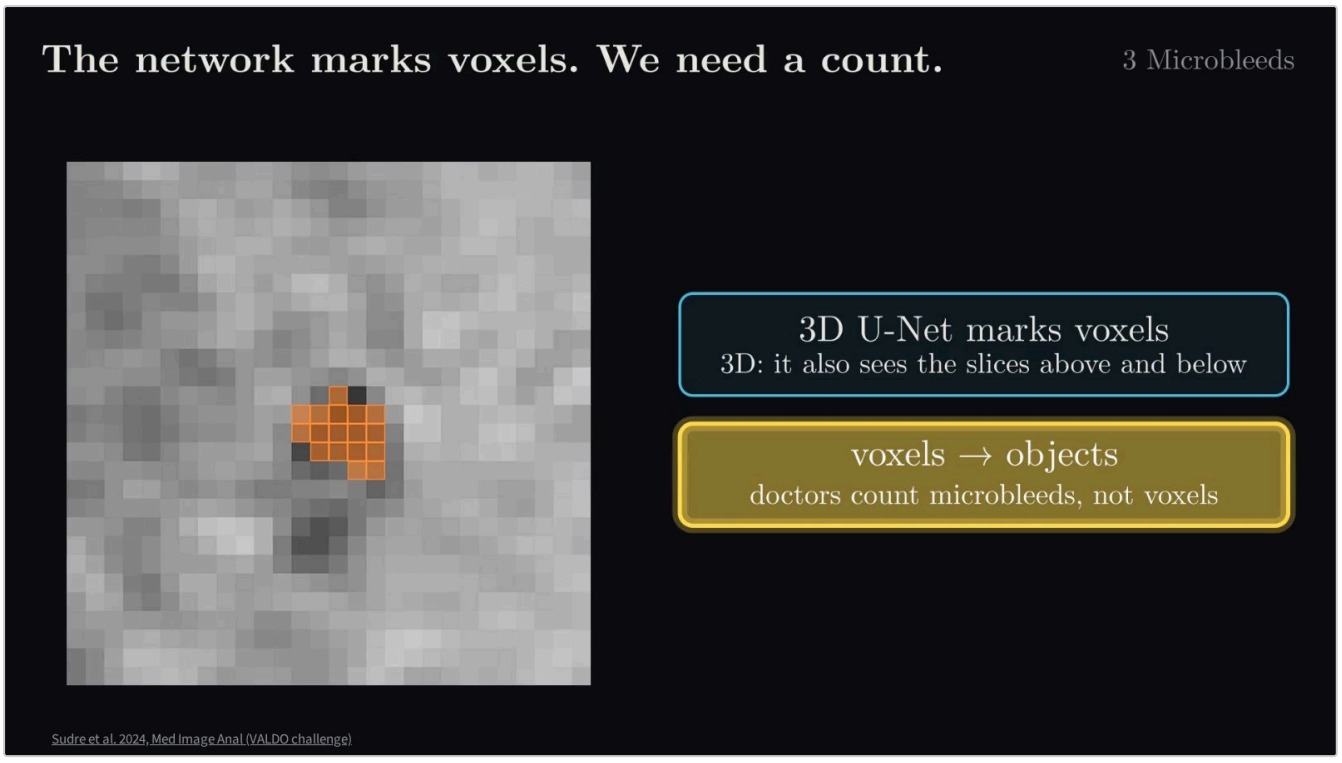
Why count them? First: microbleeds are common. In one large study, every sixth person in their sixties had some, and more than every third over eighty. Where they sit hints at the cause. Near the surface of the brain: amyloid in the vessel walls. Deep in the brain: high blood pressure. And they matter for treatment. On blood thinners, patients with microbleeds had almost four times as many brain bleeds. Still, the studies say: do not stop the drug for that reason alone. Even with ten or more microbleeds, a new stroke from a blocked vessel stays more likely than a bleed.

SOURCES

- Vernooij et al. 2008, Neurology (Rotterdam Scan Study)
  Rotterdam Scan Study (1,062 people, 1.5 T): microbleeds in 17.8 % at age 60-69 and 38.3 % over 80; lobar location points to amyloid angiopathy, deep location to hypertensive vessel disease.
- Charidimou et al. 2022, Lancet Neurol (Boston criteria 2.0)
  Boston criteria 2.0 for cerebral amyloid angiopathy: strictly lobar haemorrhagic lesions (bleeds, microbleeds, superficial siderosis) make the diagnosis.

- Wilson et al. 2018, Lancet Neurol (CROMIS-2)
  CROMIS-2: anticoagulated patients with atrial fibrillation after stroke or TIA; brain bleeds 9.8 (with microbleeds) vs 2.6 (without) per 1,000 patient-years.
- Wilson et al. 2019, Lancet Neurol (MICON pooled analysis)
  Pooled data of 20,322 patients after ischaemic stroke or TIA: with 10 or more microbleeds still 64 ischaemic strokes vs 27 brain bleeds per 1,000 patient-years.

39. The network marks voxels. We need a count.



KEY POINTS

- **3D U-Net marks voxels** – 3D: it also sees the slices above and below
- **voxels → objects** – doctors count microbleeds, not voxels

WHAT IS SAID

A three D U-Net looks at the scan as a block, with the slices above and below, and marks every voxel that looks like a bleed. But doctors count bleeds, not voxels. So we group the voxels into objects.

SOURCES

- Sudre et al. 2024, Med Image Anal (VALDO challenge)
  VALDO challenge (MICCAI 2021): automatic detection of small vessel lesions, including microbleeds.

40. Clusters: 26 neighbours



KEY POINTS

- **6 share a face**
- **12 share an edge**
- **8 share a corner**
- **connected voxels = 1 cluster = 1 microbleed**

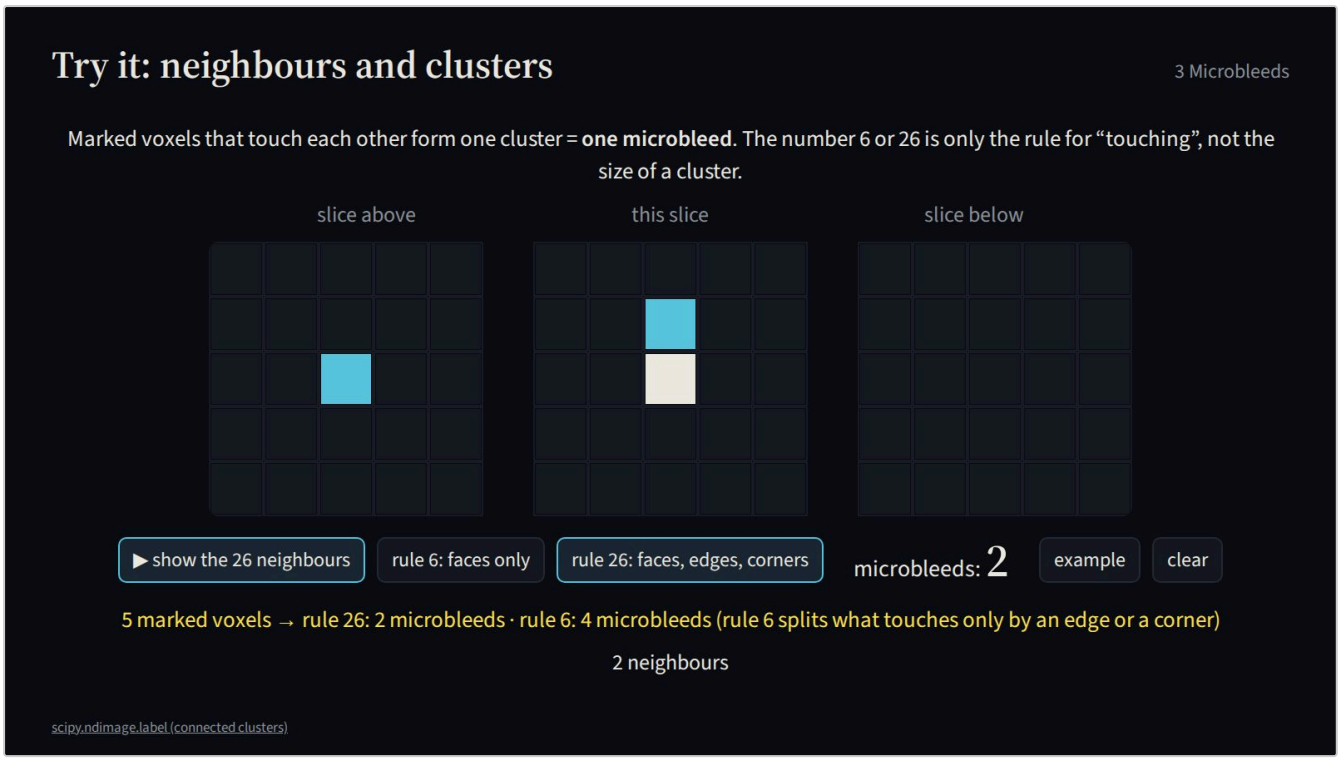
WHAT IS SAID

Voxels that touch belong together: by a face, an edge, or a corner. Twenty six neighbours. One cluster is one microbleed.

SOURCES

- scipy.ndimage.label (connected clusters)
  SciPy function that groups touching voxels into clusters; the neighbourhood (6 or 26) is a choice.

41. Try it: neighbours and clusters



> Interactive page: try it in the online deck – mendeltem.github.io/mri-segmentation-lectures/lecture_1118/deck/#try_26

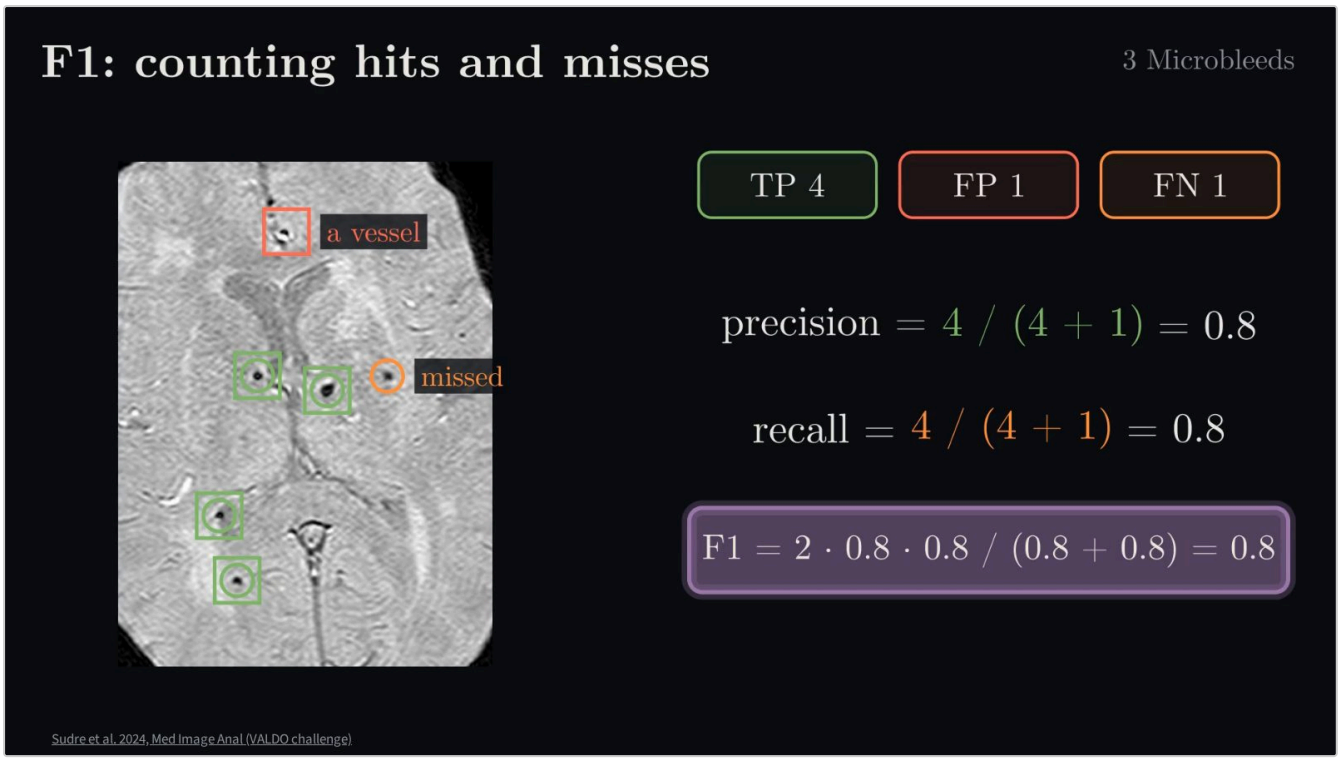
WHAT IS SAID

Click voxels on the three slices. With 26 neighbours, voxels that touch only at an edge or a corner join one cluster; with 6 they stay separate. The count of microbleeds changes with the rule.

SOURCES

- scipy.ndimage.label (connected clusters)
  SciPy function that groups touching voxels into clusters; the neighbourhood (6 or 26) is a choice.

42. F1: counting hits and misses



KEY POINTS

- **TP: found and real**
- **FP: false alarm**
- **FN: missed**
- **precision = TP / (TP + FP)**
- **recall = TP / (TP + FN)**
- **TP 4, FP 1, FN 1 → F1 0.8**

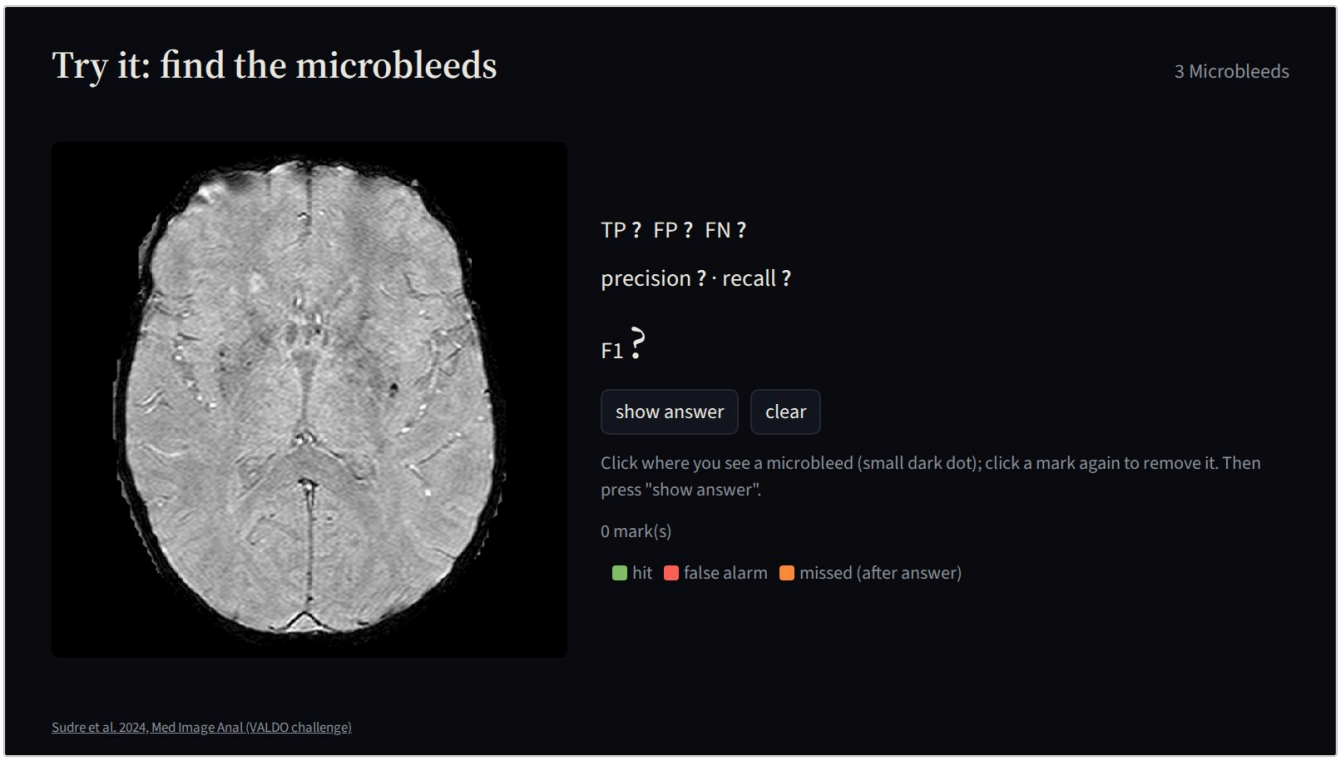
WHAT IS SAID

Now we count objects, not voxels. The expert marked five microbleeds: the circles. The program marked five places: the squares. Four are found. One is a false alarm: a vessel. And one bleed is missed. Precision: how many marks are real. Recall: how many bleeds we found. F1 combines both. Four found, one false, one missed: F1 is zero point eight.

SOURCES

- Sudre et al. 2024, Med Image Anal (VALDO challenge)
  VALDO challenge (MICCAI 2021): automatic detection of small vessel lesions, including microbleeds.

43. Try it: find the microbleeds



> Interactive page: try it in the online deck – mendeltem.github.io/mri-segmentation-lectures/lecture_1118/deck/#try_f1

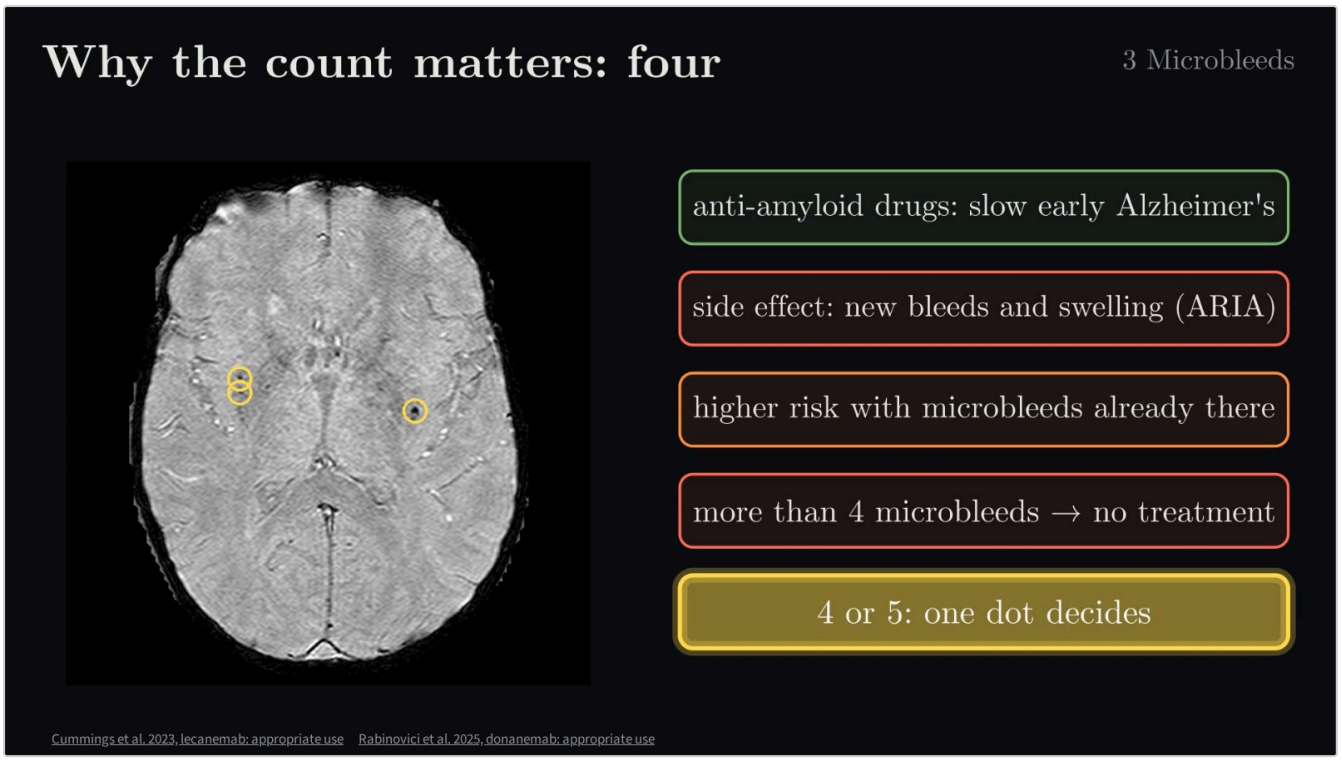
WHAT IS SAID

Click where you see a microbleed. Green: hit. Red: false alarm. 'Show answer' marks the expert's microbleeds and the misses. Precision, recall and F1 update with every click.

SOURCES

- Sudre et al. 2024, Med Image Anal (VALDO challenge)
  VALDO challenge (MICCAI 2021): automatic detection of small vessel lesions, including microbleeds.

44. Why the count matters: four



KEY POINTS

- **anti-amyloid drugs: slow early Alzheimer's**
- **side effect: new bleeds and swelling (ARIA)**
- **higher risk with microbleeds already there**
- **more than 4 microbleeds → no treatment**
- **4 or 5: one dot decides**

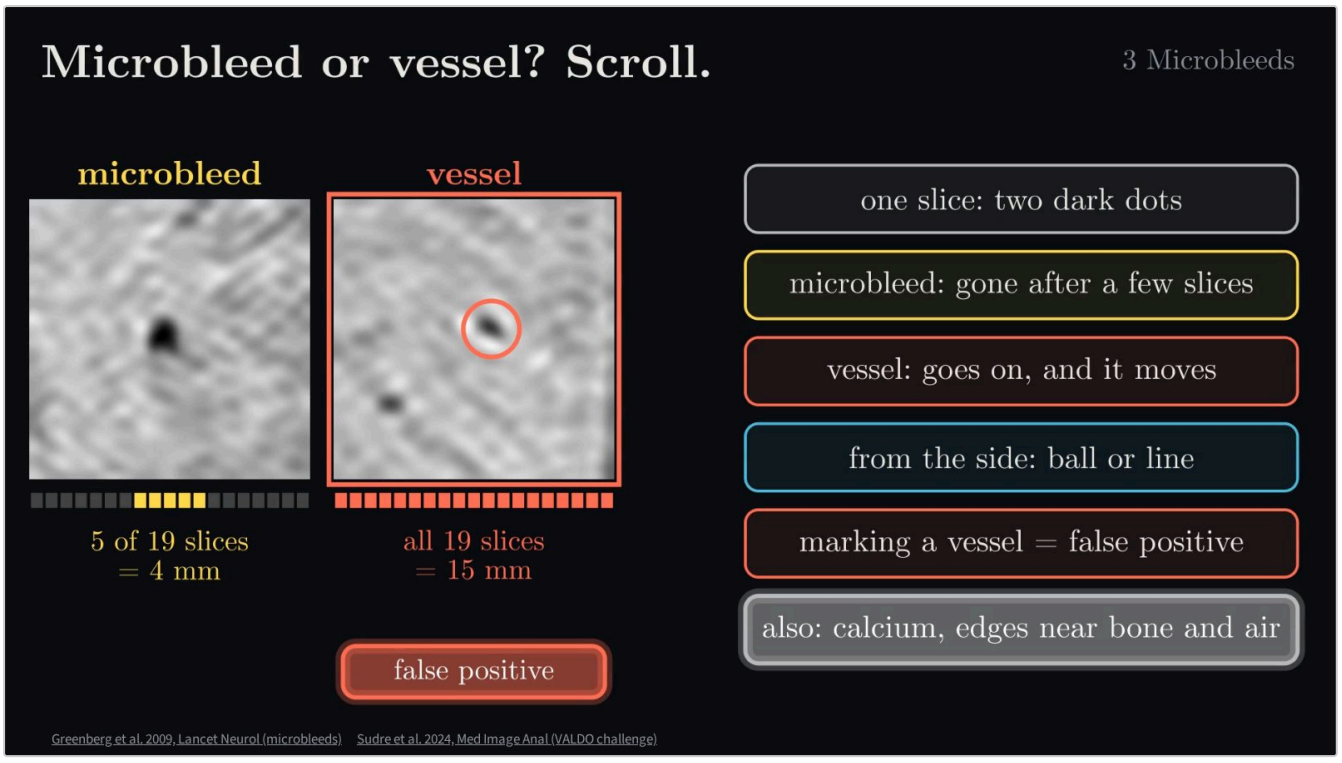
WHAT IS SAID

Why count so carefully? New Alzheimer drugs remove amyloid, but they can cause swelling and new bleeds in the brain. The risk is higher in people who already have microbleeds. So with more than four, the drug is not given. Four or five: one dot decides.

SOURCES

- Cummings et al. 2023, lecanemab: appropriate use
  Appropriate use recommendations for lecanemab: no treatment with more than 4 microhaemorrhages.
- Rabinovici et al. 2025, donanemab: appropriate use
  Appropriate use recommendations for donanemab: same limit of more than 4 microbleeds; superficial siderosis excludes.

45. Microbleed or vessel? Scroll.



KEY POINTS

- **one slice: two dark dots**
- **microbleed: gone after a few slices**
- **vessel: goes on, and it moves**
- **from the side: ball or line**
- **marking a vessel = false positive**
- **also: calcium, edges near bone and air**

WHAT IS SAID

Is every dark dot a microbleed? No. On a single slice, a microbleed and a vessel cut across look the same. So scroll through the slices. The microbleed is gone after a few slices. The vessel goes on, slice after slice, and it moves. From the side, the microbleed is a ball, and the vessel is a line. Mark the vessel, and you have a false positive. Calcium, and the edges near bone and air, can fool us the same way.

SOURCES

- Greenberg et al. 2009, Lancet Neurol (microbleeds)
  Review of cerebral microbleeds: definition, detection on T2*/SWI, mimics and clinical meaning.
- Sudre et al. 2024, Med Image Anal (VALDO challenge)
  VALDO challenge (MICCAI 2021): automatic detection of small vessel lesions, including microbleeds.

46. Your turn: find the microbleeds



> **Exercise: Microbleed exercise** (9 min) – 3 scans. Click every microbleed, then get your F1.
> mendeltem.github.io/mri-segmentation-lectures/lecture_1118/exercises/cmb-f1/

KEY POINTS

- **Microbleed exercise: 3 scans**
- **click every microbleed → your F1**

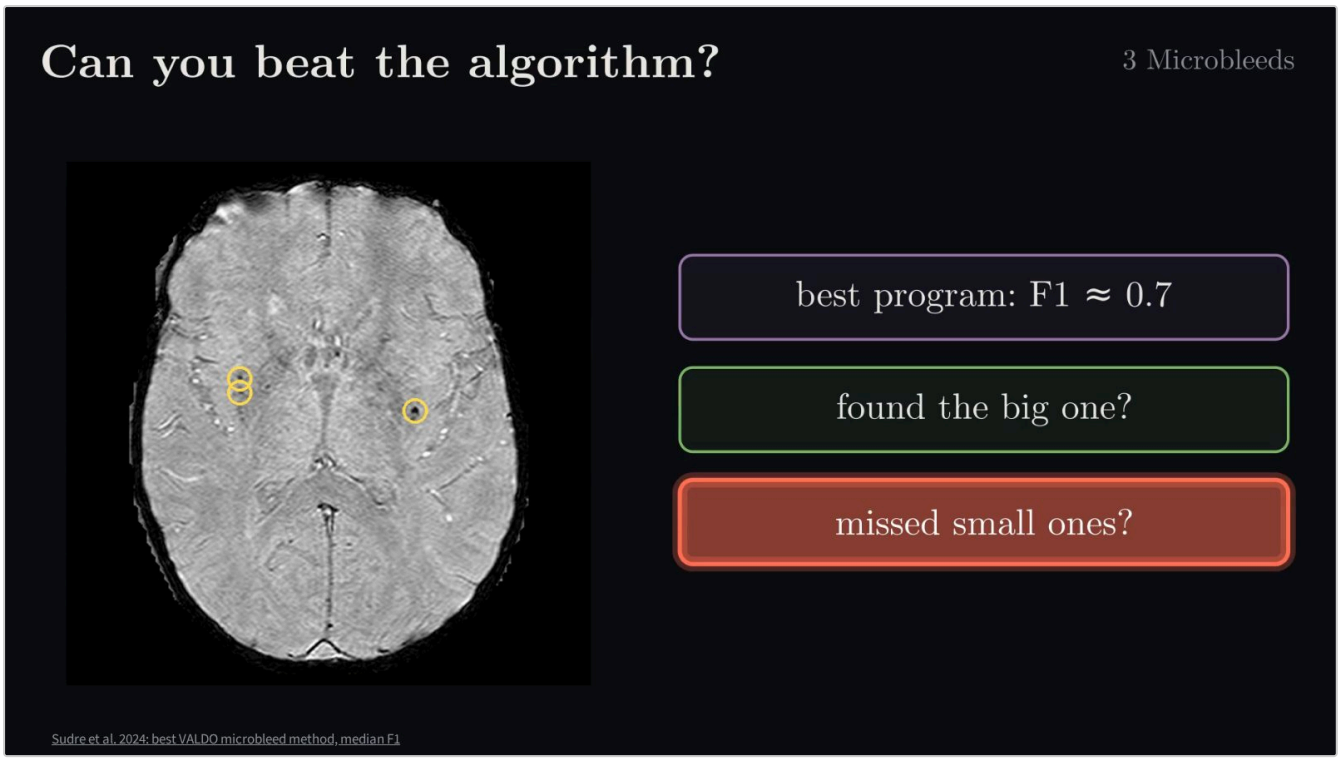
WHAT IS SAID

Last exercise. Three scans. Scroll through the slices and click on every microbleed you find. Then you get your precision, your recall and your F1. You have nine minutes.

SOURCES

- Sudre et al. 2024, Med Image Anal (VALDO challenge)
  VALDO challenge (MICCAI 2021): automatic detection of small vessel lesions, including microbleeds.

47. Can you beat the algorithm?



KEY POINTS

- **best program: F1 ≈ 0.7**
- **found the big one?**
- **missed small ones?**

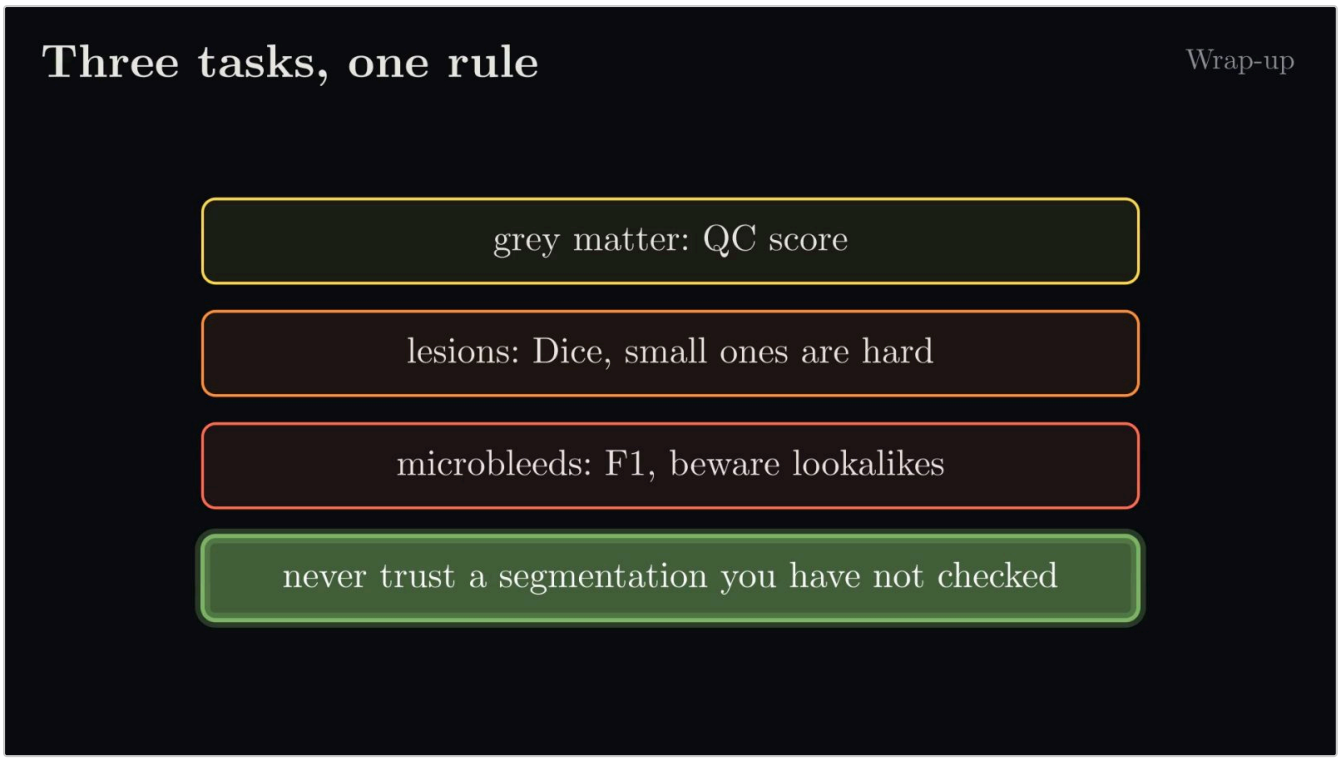
WHAT IS SAID

The best program reached an F1 of about zero point seven. Hands up: who did better? Who found the big one, and who missed small ones?

SOURCES

- Sudre et al. 2024: best VALDO microbleed method, median F1
  VALDO results: the best microbleed method reached a median F1 of about 0.68 per scan.

48. Three tasks, one rule



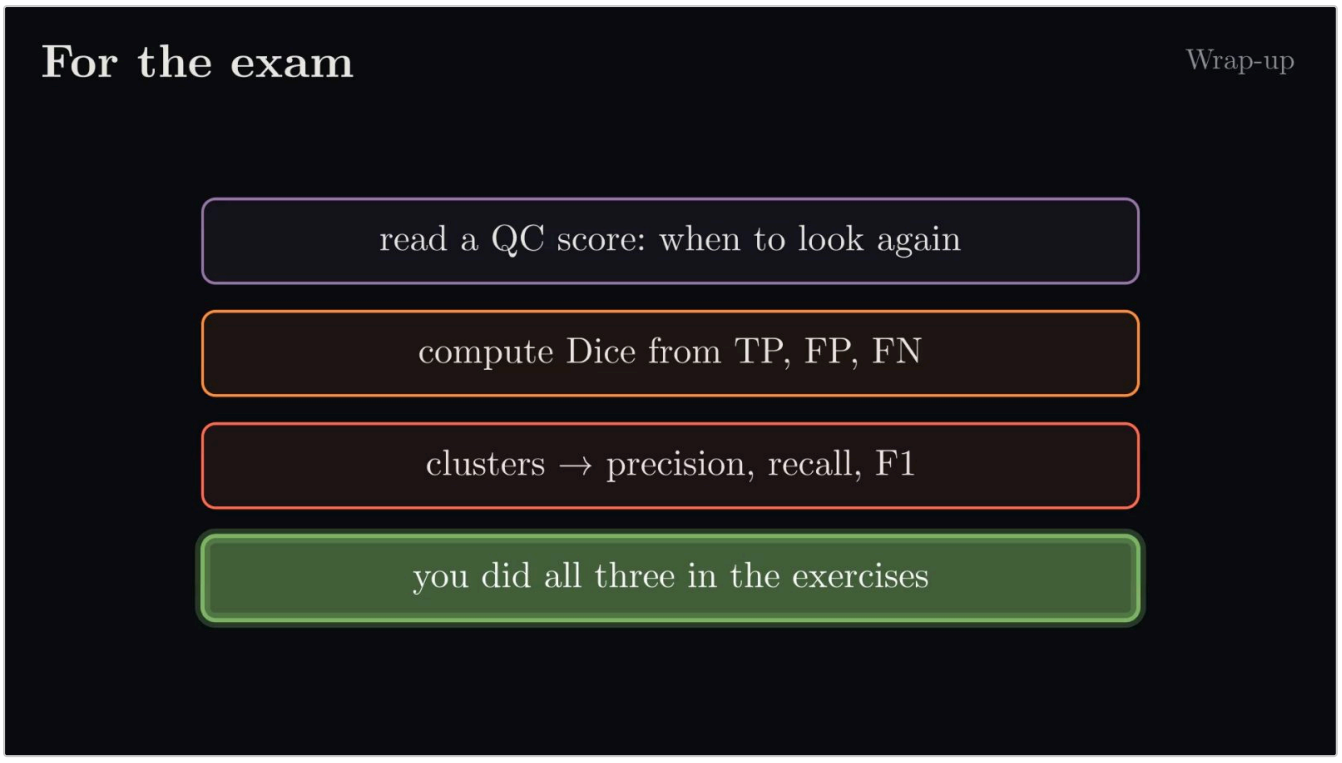
KEY POINTS

- **grey matter: QC score**
- **lesions: Dice, small ones are hard**
- **microbleeds: F1, beware lookalikes**
- **never trust a segmentation you have not checked**

WHAT IS SAID

Grey matter: check the quality score. Lesions: Dice, and small ones are hard. Micro-bleeds: F1, and beware lookalikes. One rule for all: never trust a segmentation you have not checked.

49. For the exam



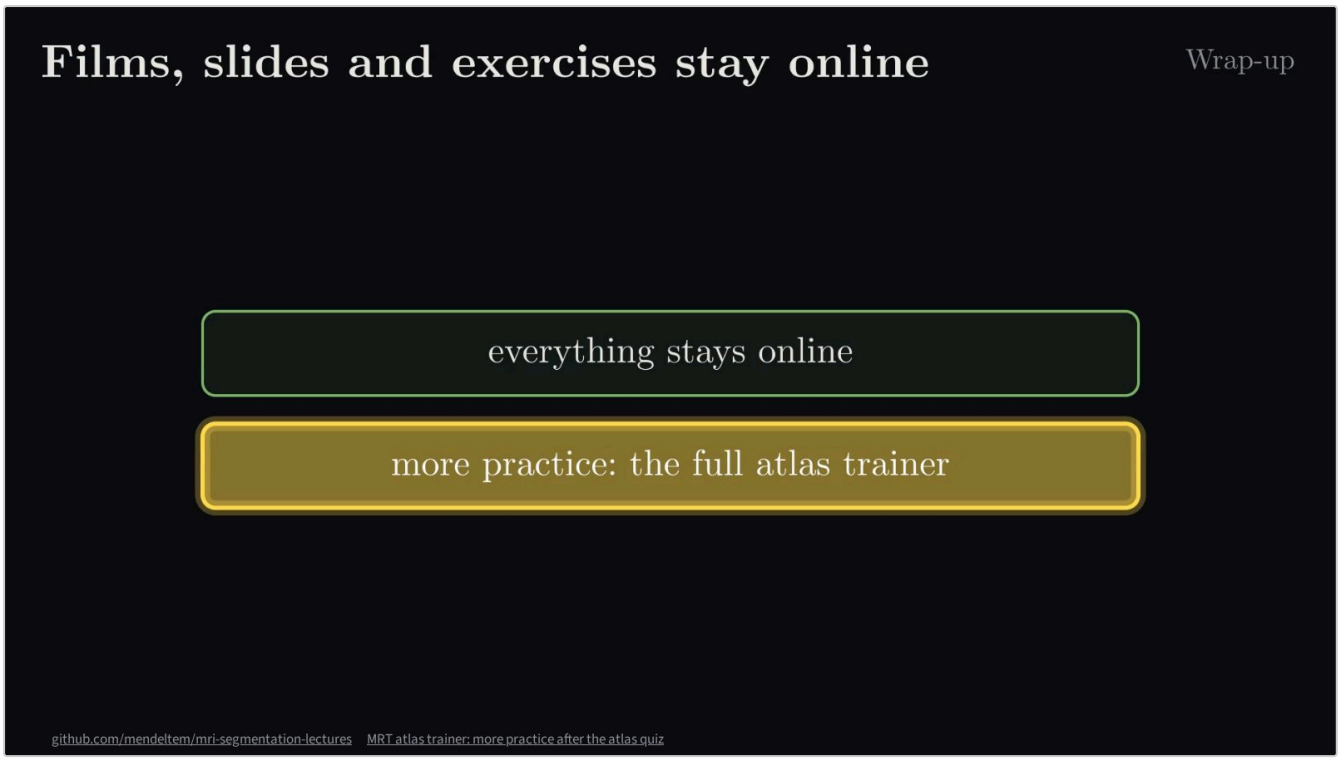
KEY POINTS

- **read a QC score: when to look again**
- **compute Dice from TP, FP, FN**
- **clusters → precision, recall, F1**
- **you did all three in the exercises**

WHAT IS SAID

For the exam: read a quality score, compute a Dice, and count clusters for an F1. You just did all three.

50. Films, slides and exercises stay online



KEY POINTS

- **everything stays online**
- **more practice: the full atlas trainer**

WHAT IS SAID

Everything stays online. For more practice, there is the full atlas trainer. Thank you.

SOURCES

- github.com/mendeltem/mri-segmentation-lectures
  The lecture's GitHub repository with slides, films and exercises.
- MRT atlas trainer: more practice after the atlas quiz
  The full atlas trainer for home: explore, quiz and study modes with many more brain regions (interface in German).

All sources

1. Battaglini et al. 2012, Hum Brain Mapp (lesion filling) – slide 18
   Lesions bias tissue segmentation; filling them with normal-looking intensities removes most of the bias.
   https://doi.org/10.1002/hbm.21344

2. Bethlehem et al. 2022, Nature (brain charts) – slide 3
   Brain charts for the human lifespan: 123,984 scans of 101,457 people, from 115 days after conception to 100 years; gives centiles like a growth chart.
   https://doi.org/10.1038/s41586-022-04554-y

3. Billot et al. 2023, Med Image Anal (SynthSeg) – slide 9, 10, 11, 12
   SynthSeg: a network trained only on synthetic images; segments scans of any contrast and resolution.
   https://doi.org/10.1016/j.media.2023.102789

4. Billot et al. 2023, PNAS (robust SynthSeg, QC) – slide 13, 14
   SynthSeg+ (robust version) with automatic QC scores per region; below 0.65 a region counts as failed.
   https://doi.org/10.1073/pnas.2216399120

5. Charidimou et al. 2022, Lancet Neurol (Boston criteria 2.0) – slide 38
   Boston criteria 2.0 for cerebral amyloid angiopathy: strictly lobar haemorrhagic lesions (bleeds, microbleeds, superficial siderosis) make the diagnosis.
   https://doi.org/10.1016/S1474-4422(22)00208-3

6. Cummings et al. 2023, lecanemab: appropriate use – slide 21, 44
   Appropriate use recommendations for lecanemab: no treatment with more than 4 microhaemorrhages.
   https://doi.org/10.14283/jpad.2023.30

7. Dale, Fischl, Sereno 1999, NeuroImage – slide 4, 7
   FreeSurfer part I: segmentation and the white and pial surface of the cortex.
   https://doi.org/10.1006/nimg.1998.0395

8. Data: Aphasia Recovery Cohort, CC0 – slide 14, 17
   Aphasia Recovery Cohort: public stroke MRI with expert lesion masks (CC0).
   https://openneuro.org/datasets/ds004884

9. Data: Dallas Lifespan Brain Study, CC0 – slide 2, 13, 17
   Dallas Lifespan Brain Study: public MRI data (CC0), used for our grey-matter pictures.
   https://doi.org/10.18112/openneuro.ds004856.v1.3.0

10. Data: WMH Segmentation Challenge, CC BY-NC 4.0 – slide 18, 20, 22, 24, 25, 26, 27, 28, 31, 35
    The public WMH Challenge data (DataverseNL), licence CC BY-NC 4.0; used in the lesion exercise.
    https://doi.org/10.34894/AECRSD

11. Debette & Markus 2010, BMJ – slide 20
    Meta-analysis: white matter hyperintensities raise the risk of stroke (about 3x) and dementia (about 2x).
    https://doi.org/10.1136/bmj.c3666

12. Desikan et al. 2006, NeuroImage – slide 6
    Automatic labelling of 34 cortical regions per hemisphere, learned from 40 hand-labelled brains.
    https://doi.org/10.1016/j.neuroimage.2006.01.021

13. Dice 1945, Ecology – slide 31
    Dice's 1945 overlap coefficient, first used to measure how often two species occur together.
    https://doi.org/10.2307/1932409

14. Fischl, Dale 2000, PNAS (cortical thickness) – slide 5
How FreeSurfer measures cortical thickness from the two surfaces; differences between people below 0.5 mm.
https://doi.org/10.1073/pnas.200033797

15. Fischl, Sereno, Dale 1999, NeuroImage – slide 6
FreeSurfer part II: the cortex inflated and mapped to a sphere; folding patterns aligned between brains.
https://doi.org/10.1006/nimg.1998.0396

16. FreeSurfer: recon-all – slide 5, 7, 8, 12
FreeSurfer documentation of recon-all, the command that runs the whole pipeline.
https://surfer.nmr.mgh.harvard.edu/fswiki/recon-all

17. github.com/mendeltem/mri-segmentation-lectures – slide 50
The lecture's GitHub repository with slides, films and exercises.
https://github.com/mendeltem/mri-segmentation-lectures

18. Gouw et al. 2008, Cerebrovasc Dis (visual scales vs volume) – slide 21
Measuring the change of lesion volume was reliable (ICC 0.88); judging progression with visual scales was less reliable (20 scan pairs).
https://doi.org/10.1159/000113863

19. Grabner et al. 2006, MICCAI (MNI152 standard brain) – slide 26
The MNI152 standard brain (average of 152 healthy brains): scans are aligned to it, so every voxel gets a position that means the same place in every person.
https://doi.org/10.1007/11866763_8

20. Greenberg et al. 2009, Lancet Neurol (microbleeds) – slide 37, 45
Review of cerebral microbleeds: definition, detection on T2*/SWI, mimics and clinical meaning.
https://doi.org/10.1016/S1474-4422(09)70013-4

21. Griffanti et al. 2016, NeuroImage (BIANCA) – slide 23, 24, 25, 26, 27, 30, 36
BIANCA: k-nearest-neighbour WMH segmentation in FSL; Dice 0.76 in one cohort, 0.52 in another.
https://doi.org/10.1016/j.neuroimage.2016.07.018

22. Inzitari et al. 2009, BMJ (LADIS study) – slide 21
LADIS: older people living independently, 3 years follow-up; disability or death per year 10.5 % (mild), 15.1 % (moderate), 29.5 % (severe white matter changes).
https://doi.org/10.1136/bmj.b2477

23. Jack et al. 2018, Alzheimers Dement (AT(N) research framework) – slide 3
Research definition of Alzheimer's disease by biomarkers: amyloid (A), tau (T), neurodegeneration (N). Atrophy on MRI is one (N) marker; not specific, and not meant for routine practice.
https://doi.org/10.1016/j.jalz.2018.02.018

24. Kuijf et al. 2019, IEEE TMI (WMH Challenge) – slide 22, 29, 30, 32, 36
WMH Segmentation Challenge: 60 training and 110 test scans, 5 scanners; best method Dice 0.80, second experts 0.77-0.79.
https://doi.org/10.1109/TMI.2019.2905770

25. MRT atlas trainer: more practice after the atlas quiz – slide 19, 50
The full atlas trainer for home: explore, quiz and study modes with many more brain regions (interface in German).
https://mendeltem.github.io/mrt-atlas-trainer/

26. Rabinovici et al. 2025, donanemab: appropriate use – slide 44
Appropriate use recommendations for donanemab: same limit of more than 4 microbleeds; superficial siderosis excludes.
https://doi.org/10.1016/j.tjpad.2025.100150

27. Ronneberger et al. 2015, MICCAI (U-Net) – slide 29
U-Net: the encoder-decoder network with skip connections for biomedical image segmentation.
https://doi.org/10.1007/978-3-319-24574-4_28

28. Rosen et al. 2018, NeuroImage (Euler number) – slide 15
The Euler number of FreeSurfer's surface flags unusable scans very well (AUC 0.98-0.99).
https://doi.org/10.1016/j.neuroimage.2017.12.059

29. Schmaal et al. 2016, Mol Psychiatry (ENIGMA depression) – slide 3
ENIGMA consortium: 1,728 patients with depression and 7,199 controls from 15 samples; hippocampus 1.24 % smaller (Cohen's d = -0.14).
https://doi.org/10.1038/mp.2015.69

30. scipy.ndimage.label (connected clusters) – slide 40, 41
SciPy function that groups touching voxels into clusters; the neighbourhood (6 or 26) is a choice.
https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.label.html

31. Sormani et al. 2014, Ann Neurol (atrophy in MS trials) – slide 3
Meta-analysis of 13 multiple sclerosis trials (more than 13,500 patients): the more a drug slowed brain atrophy, the more it slowed disability (R2 = 0.48).
https://doi.org/10.1002/ana.24018

32. SPRINT MIND 2019, JAMA (blood pressure and white matter lesions) – slide 21
SPRINT MIND: intensive vs standard blood pressure control; lesion volume 4.57 to 5.49 cm3 vs 4.40 to 5.85 cm3 after about 4 years (difference -0.54 cm3, small).
https://doi.org/10.1001/jama.2019.10551

33. Sudre et al. 2024, Med Image Anal (VALDO challenge) – slide 37, 39, 42, 43, 45, 46, 47
VALDO challenge (MICCAI 2021): automatic detection of small vessel lesions, including microbleeds.
https://doi.org/10.1016/j.media.2023.103029

34. Vernooij et al. 2008, Neurology (Rotterdam Scan Study) – slide 38
Rotterdam Scan Study (1,062 people, 1.5 T): microbleeds in 17.8 % at age 60-69 and 38.3 % over 80; lobar location points to amyloid angiopathy, deep location to hypertensive vessel disease.
https://doi.org/10.1212/01.wnl.0000307750.41970.d9

35. Wardlaw et al. 2013, Lancet Neurol (STRIVE) – slide 20
STRIVE consensus: names, definitions and imaging standards for small vessel disease (WMH, microbleeds).
https://doi.org/10.1016/S1474-4422(13)70124-8

36. Wilson et al. 2018, Lancet Neurol (CROMIS-2) – slide 38
CROMIS-2: anticoagulated patients with atrial fibrillation after stroke or TIA; brain bleeds 9.8 (with microbleeds) vs 2.6 (without) per 1,000 patient-years.
https://doi.org/10.1016/S1474-4422(18)30145-5

37. Wilson et al. 2019, Lancet Neurol (MICON pooled analysis) – slide 38
Pooled data of 20,322 patients after ischaemic stroke or TIA: with 10 or more microbleeds still 64 ischaemic strokes vs 27 brain bleeds per 1,000 patient-years.
https://doi.org/10.1016/S1474-4422(19)30197-8